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Modeling the Uptake of Antibodies Targeting the Transferrin Receptor through the Blood Brain Barrier with Ordinary Differential Equations

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Abstract

This thesis is about mathematical modeling in pharmacokinetics. In particular different approaches to model the uptake of antibodies, which are capable of binding to the transferrin receptor, through the blood brain barrier (BBB) are discussed. The dynamics are described with ordinary differential equations. Multiple simulations are calculated and important properties are proven. Different approaches to simplify the models are discussed and evaluated. One model describes the effects of bispecific antibodies on amyloid-β peptide concentration in the brain and on reticulocyte concentration, and another model describes the dynamics of monoclonal antibodies uptake through the BBB in detail.

Eidesstattliche Erklärung

Hiermit versichere ich, dass die vorliegende Masterarbeit ohne die Hilfe Dritter und nur mit den angegebenen Quellen angefertigt wurde. Alle verwendeten Passagen wurden kenntlich gemacht. Diese Masterarbeit hat in gleicher oder ähnlicher Form noch keiner Prüfungsbehörde vorgelegen.

Fabian Gottfried

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München, den 15.07.2026

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1 Introduction

During drug developed the properties and effects of a drug are tested in vitro and in vivo (with animal experiments). But those resources are limited. Applied mathematics provide a framework, how these properties and effects can be studied, once these biological systems are translated into a mathematical framework. This can help to improve the process of drug developed, to reduce the need of real world experiments and to translate result to different frameworks. (e.g. from animal to human) General speaking it helps us to understand the biological process better. Pharmacokinetics (PK) investigate the effects and the dynamics of a drug in different compartments in the body. These dynamics can be described with differential equations. (book by Caspar and Hungerbühler page 225 and following [3], 2022)

In the following chapters we study the uptake of drugs through the blood brain barrier (BBB). This is especially interesting since it is difficult to develop drugs capable of passing through the BBB. The BBB exists to protect the nerve cells in the brain by separating the blood and the brain. The BBB is selectively permeable, so that some substances can pass, which are considered necessary for the brain functions like nutrients (e.g. glucose), and others get "rejected", which are potential threatening. However, this system does not always work perfectly. Some harmful substances like alcohol can pass the BBB, and on the other hand the BBB makes it difficult for drugs to pass, which are needed in the brain to treat deceases like Alzheimer's disease or brain cancer. (AOK Gesundheitsmagazin, 2025 [6])

After providing the biological, mathematical and computational foundations we investigate the general dynamics of two simple PK models (chapter 5). The models in chapter 6 and chapter 7 show the dynamics of two antibodies (drugs) taking advantage of binding to the transferrin receptor (TfR) to pass the BBB. The model in chapter 6 focuses on the effects of the antibodies in the plasma and the brain: the unwanted side effect of reducing the reticulocyte concentration in the blood and the wanted effect of reducing the amyloid-β concentration in the brain. The model in chapter 7 focuses on the dynamics in the BBB, which consist of the luminal membrane and the endothelial intracellular compartment. In this model we describe in detail, the concentrations of the bounded and unbounded antibodies in the different compartments as well as the concentrations of the bounded and unbounded transferrin. This makes the model more complex, why we use different approaches to simplify the model.

2 Biological Foundations

Small molecules are more likely to pass the BBB, but according to the article by Grimm et al. 2023 [7], witch investigates treatments for the Alzheimer's decease, small molecule treatment have only reduced the symptoms, but to get an effect on the decease course more complex protein molecules are necessary. To help therapeutic antibodies to cross the BBB transmembrane proteins, witch can bind the antibodies are needed. Examples for these proteins are discussed, and their behavior is described with ODEs in the following chapters.

The drugs under investigation are in early states of development. This means the data provided in the literature is from in vitro testing and animal testing (in vivo). (Gunn et al., 2012 [**PET**]) During animal testing the plasma is generated out of blood samples. Then the plasma concentration of the test compound is measured with LC-MS/MS (liquid chromatography/mass spectroscopy) analysis. (Di and Kerns, 2015 [5] on page 18) The BPR (blood plasma ratio) is assumed to be a constant. That is why we model the plasma concentration, not the blood concentration of the drug. For measuring the concentration in brain tissue is taken from the animals and analyzed with LC-MS/MS. (Di and Kerns, 2015 [5] on page 19). According to the "free drug hypothesis" only the unbounded (not bounded to proteins) drug molecules are available to bind the target to produce efficacy. (Di and Kerns, 2015 [5] on page 2) This means that all models following, if not further specified, investigate the unbounded drug concentration in blood plasma (global variable $A(t)$) and the unbounded drug concentration in brain (global variable $H(t)$).

3 Mathematical Foundations

This chapter is based on the lecture notes from the ODE lecture at LMU in 2022 [12]. The definitions and the theorem can also be found in the book Gewöhnliche Differentialgleichungen by Arnold, 2001 [2].

Definition 1 (Time Derivative)
$\dot{y} = \frac{dy}{dt}$ *for* $y(t) \to \mathbb{R}^d$, $y \in C^1$, *time* $t \in \mathbb{R}$ *and* $d \in \mathbb{N}$.

Definition 2 (Initiul Value Problem)
Let us define $D \subseteq \mathbb{R} \times \mathbb{R}^{2d}$ *(open subset), the function* $F : D \to \mathbb{R} \times \mathbb{R}^{2d}$, $y_0 \in \mathbb{R}^d$ *with the initial value* $(t_0,\ y_0) \in D$, *dimension* $d \in \mathbb{N}$, *degree 1 (degree 1 means that f depends on the time t, y and the first derivative of y) and* $I \subseteq \mathbb{R}$ *interval.*

$\phi : I \to \mathbb{R}^d$ *solves the initial value problem* $\begin{cases} F(t, y, \dot{y}) = 0 \\ (t_0, y_0) \end{cases}$
with ϕ *being the solution of* $F(t, y, \dot{y}) = 0$ *under the conditions* $t_0 \in I$ *and* $\phi(t_0) = y_0$.

Note: The definition could be more general for the degree $k \in \mathbb{N}$, defining the k-th derivatives, but every IVP of higher degree can be reduced to a IVP of degree 1.

Theorem 1 (Existence and Uniqueness of a Global Solution)
For $D \subseteq \mathbb{R} \times \mathbb{R}^d$ *open subset, dimension* $d \in \mathbb{N}$, $(t_0, y_0) \in D$, $f : D \to \mathbb{R}^d$ *continuous and local Lipschitz it exists one unique solution* ϕ *of the initial value problem:* $\begin{cases} \dot{y} = f(t, y) \\ (t_0, y_0) \end{cases}$ *The graph of the solution leaves every compact interval* $(t_0, y_0) \in K \subseteq D$ *in both directions.*

Note: Since the graph leaves every compact interval in both directions we can assume that the interval I in the Definition 2 is $\mathbb{R}$ under the conditions of Theorem 1.

Definition 3 (Linear Differential Equation)
For $A \in \mathbb{R}^{d \times d}$ *matrix and* $b,\ y,\ \dot{y} \in \mathbb{R}^d$

$$\dot{y} = A \cdot y + b$$

is a linear differential equation of first order.

Note: Since a linear function is continuous and local Lipschitz, the IVP: $\begin{cases} \dot{y} & = Ay + b \\ y(t_0) & = y_0 \end{cases}$
has a unique solution.

4 Computational and Numerical Foundations

We know now, if the solution of an IVP exists. But the existence of a solution does not necessary mean, that the solution can be found easily. For linear systems an explicit solution can always be found with conventional algebraic calculations, but in general solutions can only be approximated using numerical methods. For numerical calculations the package SciPy in Python is used [4]. SciPy contains the function solve_ivp. Let us have a brief look at the Numerics of solving IVPs to understand the methods the SciPy function uses. The following is based on the article by Hairer and Lubich, year unknown [8]:

4.1 The naive way to solve an IVP

The most intuitive way of solving the IVP

$$\dot{y} = f(t, y) \text{ with the initial value } (t_0, y_0) \tag{4.1}$$

is to look at the Taylor expansion of the solution at the initial value: $y(t_0 + h) = y(t_0) + h\dot{y}(t_0) + \mathcal{O}(h^2)$. For small $h > 0$ we get the following approximation for $y_1 = y(t_0 + h)$:

$$y_1 = y_0 + hf(t_0, y_0) \tag{4.2}$$

And for the n-th step:

$$y_{n+1} = y_n + hf(t_n, y_n) \tag{4.3}$$

This formula is called the Explicit Euler Method.

4.2 Stiff problems

Let us have a look at the following ODE: $\dot{y} = \lambda y$ with $\lambda \lll 0$. The solution of the equation is $y = ce^{\lambda t}$ for an initial value $y_0 \neq 0$ and $c \neq 0$. So the that $y(t) \longrightarrow 0$ holds for $t \longrightarrow \infty$.

We calculate the solution with the Explicit Euler Method using (4.2) and (4.3):

$$(4.2) \Rightarrow y_1 = y_0 + h\lambda y_0 = (1 + h\lambda)y_0 \overset{(4.3)}{\Rightarrow} y_{n+1} = (1 + h\lambda)^n y_0$$

We see that $y_{n+1} = (1 + h\lambda)^n y_0$ does not converge at all for $n \longrightarrow \infty$ and for $\lambda \lll 0$ and $y_0 \neq 0$.

We see large negative values in a linear ODE can course problems with the numerical solvers, or in general if the Jacobian matrix $\partial_y f$ has eigenvalues with large negative real parts. (f is defined as in Theorem 1) This class of problems is called stiff problems. An example for a stiff problem would be a model with a very slow and additionally a very fast chemical reaction.

4.3 More advanced ODE solvers

The following section introduces ODE solvers, which are actually used in the SciPy package.

4.3.1 Explicit Runge-Kutta methods

We recall the fundamental theorem of calculus:

$$y(t_0 + h) = y(t_0) + h \int_0^1 y'(t_0 + \theta h)\, d\theta \tag{4.4}$$

Using the definition of f:

$$y(t_0 + h) \overset{(4.1)}{=} y(t_0) + h \int_0^1 f(t_0 + \theta h, y(t_0 + \theta h))\, d\theta \tag{4.5}$$

This integral in (4.4) and (4.5) can be approximated by a quadrature formula:

$$\int_0^1 y'\, d\theta = \sum_{i=1}^{s} b_i y_i' \tag{4.6}$$

Since the integral quadrature formula needs values for y (see 4.5), these have to be approximated with a second quadrature formula:

$$y_i = y(t_0 + c_i h) \tag{4.7}$$

This results to the following formula:

$$Y_i \overset{(4.6)(4.7)}{=} y_0 + h \sum_{j=1}^{s} a_{ij} f(t_0 + c_j h, y(t_0 + c_j h)) \tag{4.8}$$

If $a_{ij} = 0$ for $j \geqslant i$, then (4.8) can be calculated explicitly. The $s \in \mathbb{N}$ is the order of the method. The SciPy solver function uses an explicit Runge Kutta method of order 5 as default.

4.3.2 Implicit methods for stiff problems

For stiff problems implicit methods will be used. The implicit Euler uses instead of (4.3):

$$y_{n+1} = y_n + hf(t_{n+1}, y_{n+1}) \tag{4.9}$$

We notice that in (4.9) the right and the left side depend on y_{n+1}. Solving this equation methods like Newtons methods can be used. Obviously this needs a lot more computational effort. The more advanced ODE solvers can also be used implicitly.

BDF (backward differentiation formula)

This method uses numerical differentiation instead of the numerical integration in methods like Runge Kutta. In BDF we calculate a polynomial $p(t)$ which passes through the n points already calculated and additionally solves the equation:

$$\dot{p}(t_{n+1}) = f(t_{n+1}, p(t_{n+1})) \tag{4.10}$$

Radau

Like in the BDF a polynomial is calculated. In the SciPy version this polynomial has the degree 3. Furthermore, the Radau method is an implicit Runge Kutta method. This means that the derivative in the formula (4.8) is calculated implicitly. We cheep in mind that the function f in this formula is the derivative of y (4.1). SciPy uses a Runge Kutta method of order five. Radau is like all implicit Runge Kutta methods A-stable. This means, that for every $\lambda \in \mathbb{C}$ with $\mathrm{Re}(\lambda) \leqslant 0$, the numerical solution y_n stays bounded for every $n \in \mathbb{N}$ and every step size $h > 0$. So Radau is a good solver for very complex IVPs. Radau is used for the numerical calculations in this thesis.

5 First Simple Models

The first model is based on the general approach of chapter 7 and chapter 8 in the book by Caspar and Hungerbühler [3] (model on page 225 and following). With general, we mean that this model is not developed for one specific drug and not only for the BBB. This means, that it is a general PK model.

5.1 Model with two compartments

The simplest model is a 2D linear model with only the uptake $k_1 > 0$ from the blood plasma (A) to the brain (H) and a rate of elimination $\mu_H > 0$. The following diagram describes the dynamics of these two components: These dynamics can be described with an initial value problem. We define $A(t)$ as the concentration of the drug in the blood plasma and $H(t)$ as the concentration in the brain. Both terms depend on time t. $\dot{A} = \frac{dA}{dt}(t)$ and $\dot{H} = \frac{dH}{dt}(t)$ are the time derivatives of the functions $A(t)$ and $H(t)$, so these terms describe how the concentrations change with time. To be biological meaningful the non-negativity of both terms $A(t) \geqslant 0$ and $H(t) \geqslant 0$ has to be guaranteed. The non-negativity will be proven later. (Theorem 2)
The dynamics shown in Figure 5.1 define the ODEs (5.1) - (5.2):

$$\dot{A} = -k_1 A \tag{5.1}$$

$$\dot{H} = +k_1 A \ - \mu_H H \tag{5.2}$$

The initial values for the model "Figure 5.1" with the ODEs (5.1)-(5.2) are given in the vector $u_0 = (D, 0)$ with $D > 0$ being the initial concentration of the drug in the plasma after the dose is taken.
For numerically calculating the solution (figure 5.2) we choose the following parameters: the time on the interval $t = [0, 1]$ and the constants $D = 1$, $k_{PB} = 5$, $k_B = 2$. Since for now we are mainly interested in the dynamics, the graphs are shown without scales.

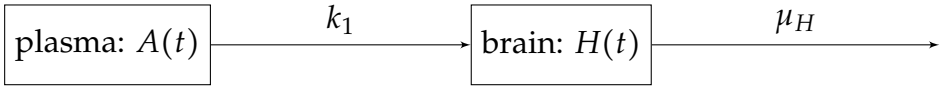


Figure 5.1: Model with the compartments: blood plasma and brain

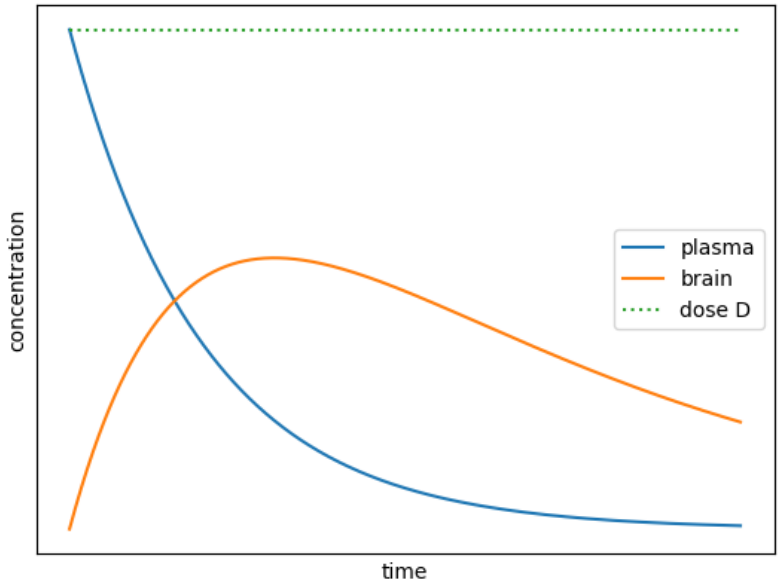


Figure 5.2: Exemplary solution of the model Figure 5.1

In the context of pharmacokinetics this results seems intuitively correct. The drug is injected in the blood, from where it is transported through the BBB to the brain. The concentration in the brain raises due to this transport to a certain level until the rate of elimination dominates, and the concentration decides in the brain.
For this easy initial value problem we can give even an explicit solution. Normally this is not reasonable, because the ODE models are more flexible and for some more complicated models, we can not give an explicit solution with mathematical terms, but we can easily calculate the solution numerically. Also, an ODE equation is often more intuitive to understand.

Theorem 2 (Solution of the linear IVP Figure 5.1)
Let $\tilde{D} := D\frac{k_1}{k_1-\mu_H}$ *for* $k_1 \neq \mu_H$
Let $\tilde{D} = 0$ *for* $k_1 = \mu_H$ *and* $k_1 > 0$, $\mu_H > 0$, $D > 0$.
Then:
The solution for the linear initial value problem with the ODEs (5.1) and (5.2) and the IV $u_0 = (D, 0)$ *is:*

$$\begin{aligned} A(t) &= D \cdot \quad e^{-k_1 t} \\ H(t) &= \tilde{D} \cdot (e^{-\mu_H t} - \quad e^{-k_1 t}) \end{aligned} \tag{5.3}$$

Proof:
Step 1: Check the initial values

$$A(0) = D \cdot 1 = D$$
$$H(0) = \tilde{D}(1-1) = 0$$

Step 2: Calculate the derivatives

$$\dot{A} = \frac{dA}{dt}(t) = -k_1 \cdot De^{-k_1 t} \overset{(5.3)}{=} -k_1 A(t) \overset{(5.1)}{=} \dot{A}$$

$$\begin{aligned}
\dot{H} &= \frac{dH}{dt}(t) \\
&= \tilde{D}(-\mu_H e^{-\mu_H t} + k_1 e^{-k_1 t}) \\
&= D \cdot \frac{k_1^2}{k_1 - \mu_H} e^{-k_1 t} - \tilde{D}\mu_H e^{-\mu_H t} \\
&= (D \cdot k_1 + D \cdot \frac{\mu_H k_1}{k_1 - \mu_H}) e^{-k_1 t} - \tilde{D} \cdot \mu_H e^{-\mu_H t} \\
&= k_1 \cdot De^{-k_1 t} - \mu_H \tilde{D}(e^{-\mu_H t} - e^{-k_1 t}) \\
&\overset{(5.3)}{=} k_1 A(t) - \mu_H H(t) \\
&\overset{(5.2)}{=} \dot{H}
\end{aligned}$$

Step 3: Non-negativity
$A(t)$ is not negative since $D > 0$ and the exponential function is always positive.
For $H(t)$:
Case 1: If $\tilde{D} = 0$, non-negativity is obviously fulfilled.
Case 2: For $\tilde{D} \neq 0$, we notice:

$$\tilde{D} < 0 \Leftrightarrow D \cdot \frac{k_1}{k_1 - \mu_H} < 0 \overset{D>0\ k_1>0}{\Longleftrightarrow} k_1 - \mu_H < 0 \Leftrightarrow k_1 < \mu_H$$

and

$$e^{-\mu_H t} - e^{-k_1 t} < 0 \overset{t>0}{\Longleftrightarrow} k_1 < \mu_H$$

This means, that the product $\tilde{D} \cdot (e^{-\mu_H t} - e^{-k_1 t}) = H(t)$ is positive.

■

Note: How to find the solution of a linear ODE with a matrix exponential is described in the book by Arnold, 2001 [2] on page 158 and the following. For the diagonalization of the matrix the matrix calculator [11] was used.

5.2 Linear Model with three compartments

The following formula is presented in the book by Di and Kerns, 2015 [5] on page 18 for the total drug concentration in blood plasma:

$$A(t) = \frac{F \cdot D}{V}(e^{-k_e \cdot t} - e^{-k_a \cdot t}) \tag{5.4}$$

with the F being the oral bioavailability, D the administered dose and V the volume distribution. The exponents are the rate of absorption k_a and the rate of elimination k_e. Since we are interested in the dynamics and not in the exact values let us modify the equation (5.4) to:

$$A(t) = K_1 e^{-k_e \cdot t} - K_2 e^{-k_a \cdot t} \tag{5.5}$$

with K_1 and K_2 being positive constants, which are not further described and are not necessary equal, making the model simpler and more flexible. We notice, that the variables (5.4) and (5.5) are more similar to the second variables of (5.3) in Theorem 2. That suggests, that the blood plasma is the second compartment for this model described by (5.5). If we assume, that the concentration in the plasma ($A(t)$) is the second variable in the system, we need to add an extra variable for the concentration in a periphery compartment (global variable $P(t)$). This makes sense for example, if the drug is taken orally meaning it has to go through the stomach before being available in the blood plasma.
The ODE system Figure 5.3 models the concentration of the drug in the three compartments **periphery** $P(t)$, **blood plasma** $A(t)$ and **brain** $H(t)$.

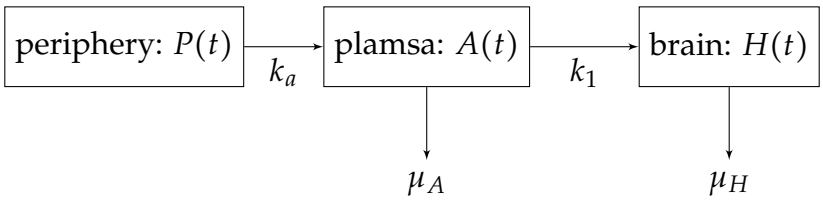


Figure 5.3: Model with the compartments: periphery, blood plasma and brain

The dynamics shown in Figure 5.3 define the ODEs (5.6) - (5.8):

$$\dot{P} = -k_a P \tag{5.6}$$
$$\dot{A} = k_a P - (k_P + k_1) A \tag{5.7}$$
$$\dot{H} = k_1 A - k_B H \tag{5.8}$$

with $k_A + k_1 = k_e$.
Next we plot the numerical solutions of the IVP for the ODE system (5.6) - (5.8). For the IVs vector $y_0 = (1, 0, 0)$ was chosen, since we assume the drug concentration is 0, before the drug is taken orally. Units are not given, because this model does not use real world data.

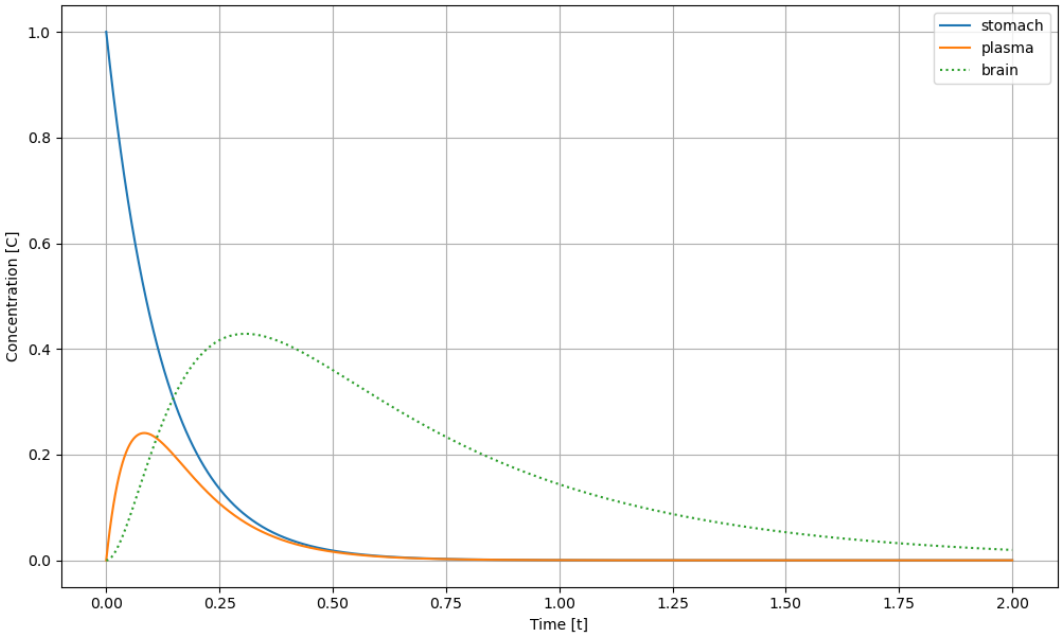


Figure 5.4: Model Figure 5.3 for k_e, k_p, k_{pb}, k_b = 8, 5, 12, 2 (the periphery compartment is labeled as stomach)

The shape of these curves for the variables give us a first impression, what we can expect for the variables of the more complex models following.

6 Modeling the Amyloid β-concentration in the Brain

This chapter is based on the models from the articles by Gadkar, Yadav, Zuchero et al. 2016 [10] and by Grimm et al. 2023 [7]. The model (equation (6.1) and following) and parameter values come from [10]. One parameter was left out for simplification.

6.1 Biological description of the compartments

The article [10] gives us a glance on the general dynamics of the protein amyloid-β. The goal of the drug is to reduce the amyloid-β concentration in the brain with bispecific antibodies like anti-BACE1. BACE1 is responsible for the cleavage of APP (amyloid precursor protein), which gives rise to amyloid β peptides, which are found in the brains of Alzheimer's disease patients. Due to the BBB the antibodies have limited access to the brain. However, studies with mice show, that binding those to TfR improves the brain uptake capacity. Our focus is to investigate the effect of "Transferrin Affinity" (K_m^{TfR}) and "Reticulocyte Affinity" (K_m^{retic}). The following model gives us insights on the effect of these parameters to the concentration of amyloid-β (function $D(t)$ (6.4)) and of reticulocyte (function $E(t)$ (6.5)).

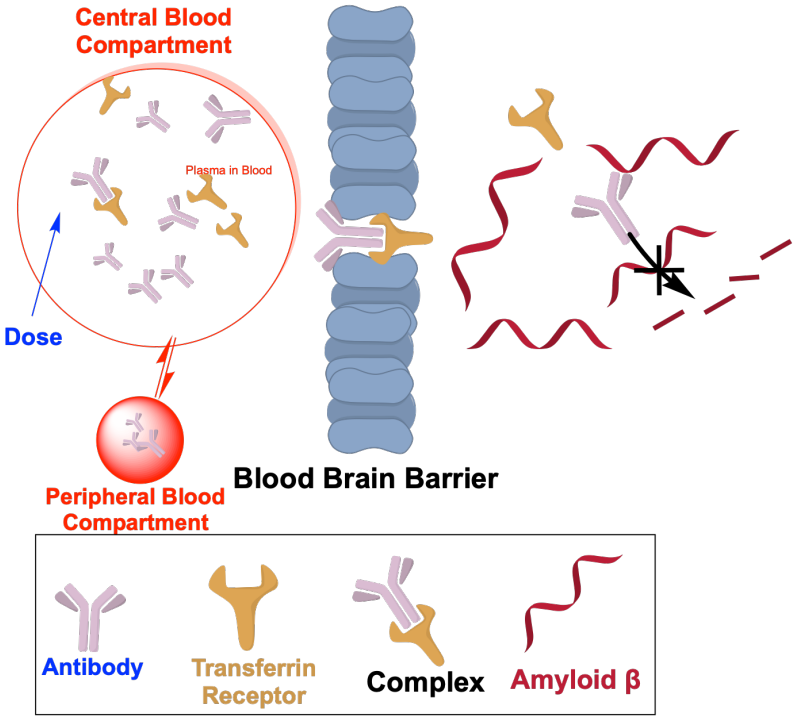


Figure 6.1: Figure shows how the antibody bounded to the transferrin receptor passes the BBB to reduce the amyloid-β concentration in the brain (blood: left, brain: right). This figure was inspired by "Figure 1" in the article by Kanodia et al. 2016 [9]

6.2 Model

This model describes the pharmacokinetics after a single intravenous drug dose given to mice. Like in the linear three-dimensional model in section 5.2, this model investigates the drug concentration in three different compartments: The "central compartment" $A(t)$, the "periphery compartment" $P(t)$ and the "brain compartment" $H(t)$. To analyze the effectiveness of the drug, we investigate the amyloid-β concentration $D(t)$, which should be reduced. To analyze the safety of the drug, we investigate the reticulocyte concentration in the brain $E(t)$, which should stay in a specific safety margin. This means that it should not decease too much.

6.2.1 The five compartments and their dependencies

This figure 6.2 shows the dynamics between the compartments:

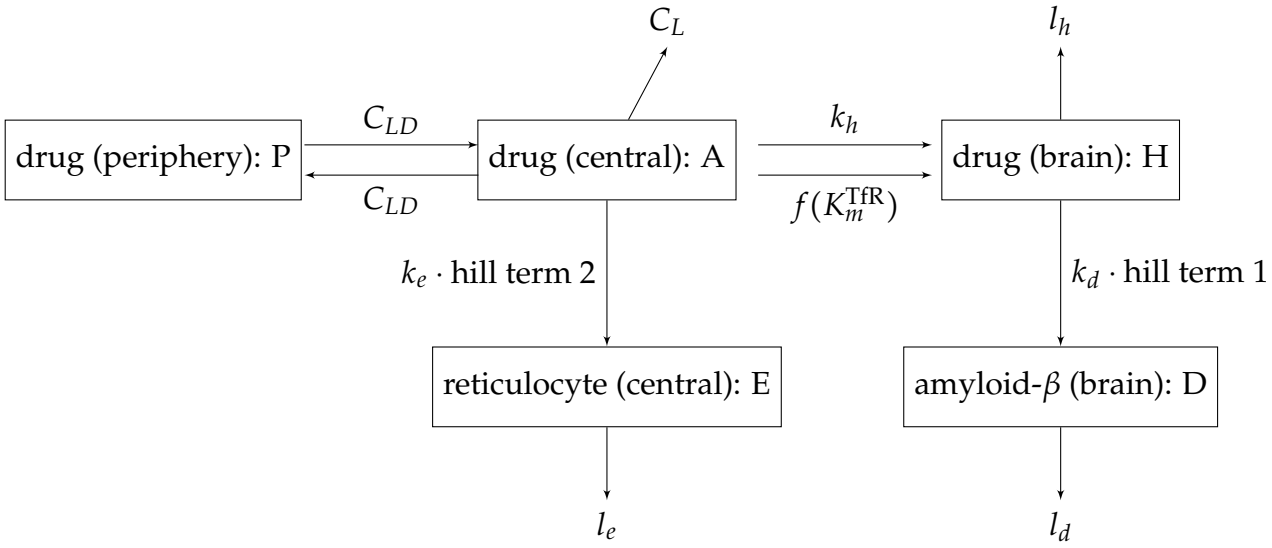


Figure 6.2: The ODE model with the boxes represent the concentrations in the compartment and the arrows their dependencies

All five terms can be defined as solutions of ODEs:

6.2.2 System of equations

$$\dot{A} = \frac{1}{V_1}\left(-C_{LD}\,(A-P) - C_L A - \frac{V_m^{\mathrm{BACE}} A}{K_m^{\mathrm{BACE}} + A} - \frac{V_m^{\mathrm{TfR}} A}{K_m^{\mathrm{TfR}} + A}\right) \tag{6.1}$$

$$\dot{P} = \frac{1}{V_2} \cdot C_{LD}\,(A-P) \tag{6.2}$$

$$\dot{H} = k_h A + f(K_m^{\mathrm{TfR}}) \cdot \frac{A}{K_m^{\mathrm{TfR}} + A} - l_h H \tag{6.3}$$

$$\dot{D} = k_d\,(1 - h_1(H)) - l_d D \tag{6.4}$$

$$\dot{E} = k_e\,(1 - h_2(A)) - l_e E \tag{6.5}$$

With:

$$f(K_m^{\mathrm{TfR}}) := \left(\alpha \log_{10}\left(K_m^{\mathrm{TfR}}\right) + \beta\right) \tag{6.6}$$

$$h_1(H) := \frac{V_m^{A\beta} \cdot H^n}{\mathrm{IC50}^n + H^n} \tag{6.7}$$

$$h_2(A) := \frac{V_m^{\mathrm{retic}} \cdot A^{n_1}}{\left(K_m^{\mathrm{retic}}\right)^{n_1} + A^{n_1}} \tag{6.8}$$

6.2.3 Definition of the Hill terms

The functions h_1 and h_2 use the logic of the hill equation: (Antwerpes 2026 [1])
Note: We defined h_1 and h_2 as functions of H and A, but in the next steps we will write $h_1(t)$ and $h_2(t)$. Since the concentration functions H and A depend on t and since we study the dynamics vs time, this is reasonable.

Definition 4 (Theta for Hill Equation)
We define the following terms:

$$\theta = \frac{C^N}{K_A^N + C^N} \qquad \textit{(general term)} \tag{6.9}$$

$$\theta_1 = \frac{H^n}{IC50^n + H^n} \qquad \textit{(θ for h_1 (6.7))} \tag{6.10}$$

$$\theta_2 = \frac{A^{n_1}}{\left(K_m^{retic}\right)^{n_1} + A^{n_1}} \qquad \textit{(θ for h_2 (6.8))} \tag{6.11}$$

θ is the share of molecules (amyloid-β or reticulocyte), which have bound to the drug. C is the drug concentration. $N > 0$ is called hill coefficient.

The fact that the hill coefficient is greater than 1 for h_1 and h_2 (n and n_1 in the data) implies, that the binding of one drug molecule to C helps a second drug molecule to bind to C. K_A is an affinity parameter. The "half maximal inhibitory concentration" IC50 qualifies the binding affinity of the drug antibodies to amyloid-β in the brain.

Lemma 1
Let the concentration C and K_A be positive.
Then: $\theta \in (0,1)$

Proof:
With these assumptions the numerator: C^N (6.9) and the denominator: $K_A^N + C^N$ (6.9) are positive, and $K_A^N + C^N > C^N$.

■

Theorem 3
For the given parameter and initial values (tables 6.4 and 6.5) and assuming the non-negative drug concentration in central compartment ($A(t) \geq 0$) we conclude for the drug concentrations in central compartment $A(t)$ and in the brain $C(t)$:

1. $A(t) \xrightarrow{t\to\infty} 0$
2. $H(t) \xrightarrow{t\to\infty} 0$

Proof: ($A(t) \overset{t\to\infty}{\longrightarrow} 0$ implies $C(t) \overset{t\to\infty}{\longrightarrow} 0$)

k_h, K_m^{TfR} and $l_h > 0$, f does not depend on $A(t)$

$\Rightarrow k_h A(t) \overset{t\to\infty}{\longrightarrow} 0$, $\dfrac{A(t)}{K_m^{\mathrm{TfR}} + A(t)} \overset{t\to\infty}{\longrightarrow} 0$

$\Rightarrow C(t) \overset{t\to\infty}{\longrightarrow} 0$

■

Proof: ($A(t) \overset{t\to\infty}{\longrightarrow} 0$)

We proof by contradiction:

assume $A(t) \overset{t\to\infty}{\longrightarrow} c > 0$ ($A(t) < 0$ is not possible)

We define $L(t) := V_1 A(t) + V_2 B(t)$

from our assumption, V_0, $V_1 > 0$ (table 6.4) and $B \geq 0$ we get $L(t) \overset{t\to\infty}{\longrightarrow} \hat{c} > 0$

$\dot{L} := V_1 \dot{A} + V_2 \dot{B} \overset{(6.1),(6.2)}{=} -C_L A - \dfrac{V_m^{\mathrm{BACE}} A}{K_m^{\mathrm{BACE}} + A} - \dfrac{V_m^{\mathrm{TfR}} A}{K_m^{\mathrm{TfR}} + A} < -C_L A < 0$

The < holds since all parameters are positive (table 6.4) and A>0

$\Longrightarrow \dot{L}(t) \overset{t\to\infty}{\longrightarrow} -C_L c < 0 \overset{L(t)\in(0,V_1 A_0) \text{ for } t>0}{\Longrightarrow} L(t) \overset{t\to\infty}{\longrightarrow} 0 \lightning$

■

Note: The initial values $A_0 > 0$ $B_0 = 0$ (table 6.5).

Theorem 4

The assumptions from theorem 3 imply:

$h_1(A) \overset{t\to\infty}{\longrightarrow} 0$ *and* $h_2(C) \overset{t\to\infty}{\longrightarrow} 0$

We know from theorem 3 $A(t) \overset{t\to\infty}{\longrightarrow} 0$ and $C(t) \overset{t\to\infty}{\longrightarrow} 0$

This implies $h_2(A) \overset{(6.8)}{=} \dfrac{V_m^{\mathrm{retic}} \cdot A^{n_1}(t)}{\left(K_m^{\mathrm{retic}}\right)^{n_1} + A^{n_1}(t)} \overset{t\to\infty}{\longrightarrow} 0$ and $h_1(C) \overset{(6.7)}{=} \dfrac{V_m^{A\beta} \cdot C^n(t)}{\mathrm{IC50}^n + C^n(t)} \overset{t\to\infty}{\longrightarrow} 0$,

if K_m^{retic} and IC50 are > 0.

■

The following figure displays the proven behavior of $h_1(t)$ (6.7) and $h_2(t)$ (6.8):

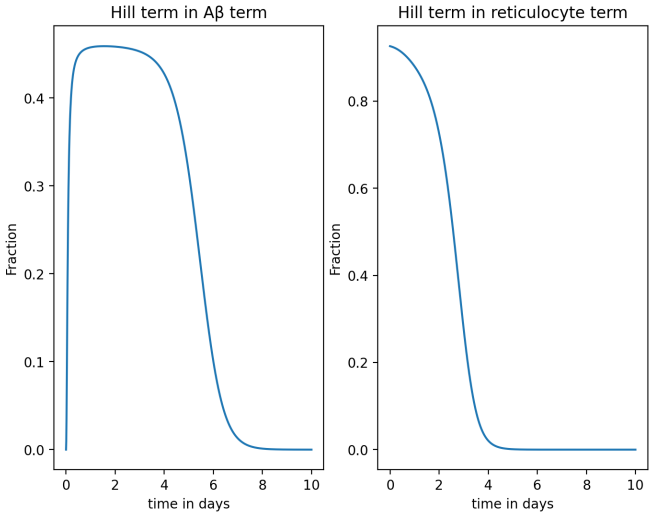


Figure 6.3: $h_1(t)$ (6.7) and $h_2(t)$ (6.8) with parameter and initial values as printed in the tables 6.4 and 6.5

We see that both terms converge to 0 as expected. At the starting time $t = 0$ the drug is injected to the central compartment. This explains why the initial concentration in the brain is 0 and positive in the central compartment at the beginning. This implies $h_1(0) = 0$ and $h_2(0) > 0$.

6.2.4 Plotting one numerical solution

The following plot shows the numerical solution of the ODE system (6.1) - (6.5) for the parameter values and IVs in the tables 6.4 and 6.5:

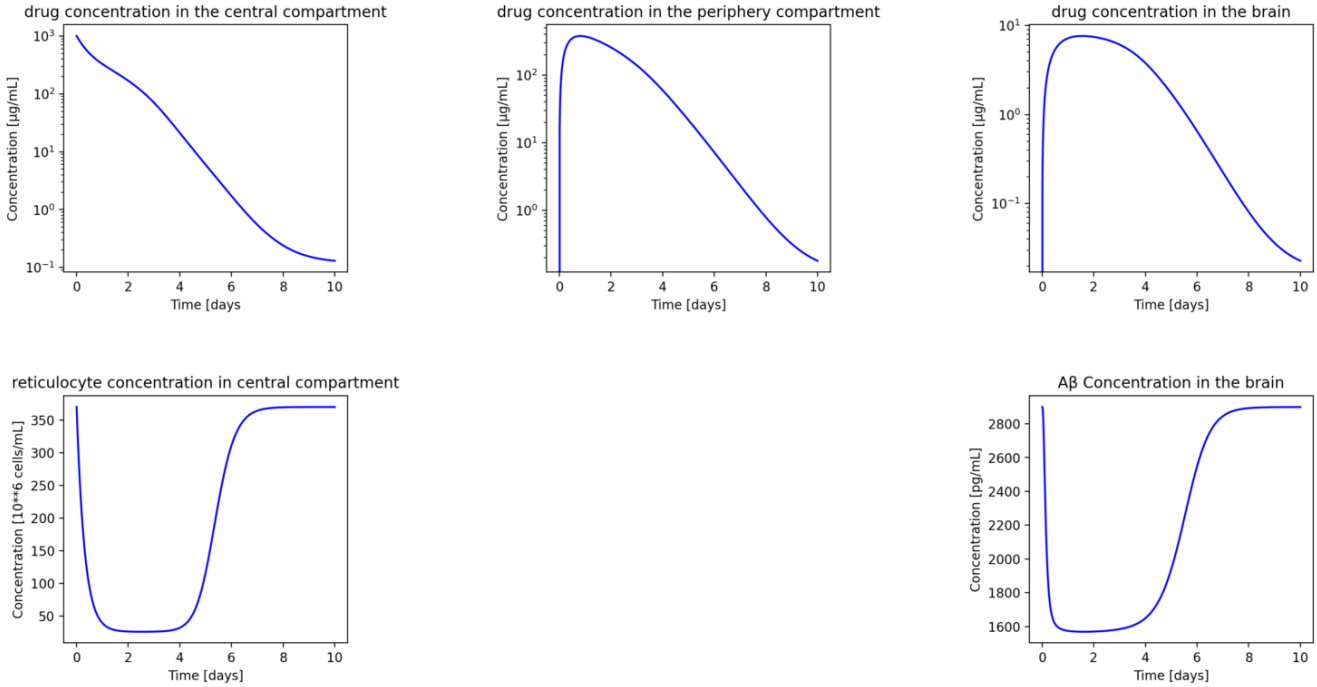


Figure 6.4: Solutions of the model (6.1) - (6.5) for the parameters and IVs in the tables 6.4 and 6.5

For the affinities we choose the parameters of antibody (drug) "A", which will be introduced soon:

abbreviation	describtion	value	unit
K_m^{TfR}	Transferrin Affinity	66.6	μg/mL
K_m^{retic}	Reticulocyte Affinity	5.32	μg/mL

Table 6.1: Affinity parameters for the figure 6.4

The concentration of the drug in the periphery and central compartment are actually solutions of the ODE model with the three equations (6.1), (6.2) and (6.3), since these equations do not depend on $D(t)$ and $E(t)$. We see similar behavior, when we compare the solution of this model to the solution of the three compartment linear model in section 5.2 (compare figure 6.4 with figure 5.4). These similarities occur, although the model in section 5.2 is not linear. In the compartment in which the drug is injected (stomach in section 5.2 and central compartment here), we see an exponential (or almost exponential) decease of concentration. In the other non-brain compartment and in the brain, we see a steep increase in both models and after the peak we see a slow decrease. The brain has the slowest decrease and the last peak in both models.

6.2.5 Dynamics of protein binding

For a better understanding of the binding process we plot the function $h_1(t)$ (6.7) vs the numerical solution of $D(t)$ (figure 6.5) and the function $h_2(t)$ (6.8) vs the numerical solution of $E(t)$ (figure 6.6). We see in the plots, that if there is a significant share of molecules being able to bind the drug, the amyloid-β and the reticulocyte concentrations drop immediately and both stay at low levels. The value of h_2 stays above a certain level until day 3, although the drug concentration in the central compartment declines. The value of h_2 stays even longer at a high level after rapidly increasing from 0. When the drug concentration falls below a certain level, the Hill terms $h_1(t)$ and $h_2(t)$ both rapidly drop to 0. We can proof this behavior with the Hill equation (6.9): Since the concentration variable C is in the numerator and in the denominator, the θ term is close to 1 for large concentrations. We propose and proof the following theorem:

Theorem 5
Characteristics of the Hill equation (6.9):

1. *For large C (concentration of the drug) and a non-zero Hill coefficient n the θ term in the Hill equation (6.9) is close to 1.*
2. *K_m^{retic} (6.8) and IC50 (6.7) represent the drug concentration, for which half of the molecules are bounded to the drug.*

Proof part 1:

$$\lim_{C\to\infty} \theta(C) \overset{(6.9)}{=} \lim_{C\to\infty} \frac{C^N}{K_A^N + C^N} \overset{L'Hospital}{=} \lim_{C\to\infty} \frac{\frac{d}{dC}C^N}{\frac{d}{dC}\left(K_A^N + C^N\right)} = \lim_{C\to\infty} \frac{N\cdot C^{N-1}}{N\cdot C^{N-1}} = 1 \text{ for } N \neq 0$$

Proof part 2:

$$\text{Let: } C = K_A \text{ ,then: } \theta \overset{(6.9)}{=} \frac{C^N}{K_A^N + C^N} \overset{C=K_A}{=} \frac{C^N}{K_A^N + C^N} = \frac{K_A^N}{K_A^N + K_A^N} = \frac{1}{2}.$$

In the next step we investigate what hilleq:general to the derivatives $\dot{D}$ and $\dot{E}$ at the boundaries $\theta = 0$ and $\theta = 1$ of the hill term (6.9). For $\dot{D}$ (6.4) we get:

Theorem 6
Let $V_m^{A\beta} \in (0,1)$, $V_m^{retic} \in (0,1)$, $\theta_1 := h_1(H)$ (6.7) and $\theta_2 := h_2(A)$ (6.8).
We find the following properties for $\dot{D}(t)$ (6.4) and $\dot{E}(t)$ (6.5) at the boundaries 0 and 1 of the Hill terms θ_1 and θ_2:

1. *$D = 0$ and $\theta_1 = 0 \implies \dot{D} = k_d$ (the concentration is rapidly increasing, if $k_d \gg 0$)*
2. *$D = \frac{k_d}{l_d}$ and $\theta_1 = 0 \implies \dot{D} = 0$ (D reaches a plateau at $D = \frac{k_d}{l_d}$)*

3. $D = 0$ *and* $\theta_1 = 1 \implies \dot{D} = k_d \left(1 - V_m^{A\beta}\right)$ *(the concentration is rapidly increasing, if* $k_d \gg 0$ *and* $V_m^{A\beta} \approx \frac{1}{2}$*)*
4. $E = 0$ *and* $\theta_2 = 0 \implies \dot{E} = k_e$ *(the concentration is rapidly increasing, if* $k_e \gg 0$*)*
5. $E = \frac{k_e}{l_e}$ *and* $\theta_2 = 0 \implies \dot{E} = 0$ *(*E *reaches a plateau at* $E = \frac{k_e}{l_e}$ *)*
6. $E = 0$ *and* $\theta_2 = 1 \implies \dot{E} = k_e \left(1 - V_m^{retic}\right)$

Proof:

$$\begin{aligned}
\dot{D} &\overset{(6.4),\,(6.10)}{=} k_d \left(1 - h_1(H)\right) - l_d D \\
&\overset{(6.7)}{=} k_d \underbrace{\left(1 - V_m^{A\beta}\theta_1\right)}_{=1 \text{ for } \theta_1 = 0} - l_d D \\
&= \begin{cases} k_d & \text{for } D = 0 \text{ and } \theta_1 = 0 \rightarrow \ 1. \\ 0 & \text{for } D = \frac{k_d}{l_d} \text{ and } \theta_1 = 0 \rightarrow \ 2. \\ k_d \left(1 - V_m^{A\beta}\right) & \text{for } D = 0 \text{ and } \theta_1 = 1 \rightarrow \ 3. \end{cases} \\
\dot{E} &\overset{(6.5),\,(6.11))}{=} k_e \left(1 - h_2(A)\right) - l_e E \\
&\overset{(6.8)}{=} k_e \underbrace{\left(1 - V_m^{\text{retic}}\theta_2\right)}_{=1 \text{ for } \theta_2 = 0} - l_e E \\
&= \begin{cases} k_e & \text{for } E = 0 \text{ and } \theta_2 = 0 \rightarrow \ 4. \\ 0 & \text{for } E = \frac{k_e}{l_e} \text{ and } \theta_2 = 0 \rightarrow \ 5. \\ k_e \left(1 - V_m^{\text{retic}}\right) & \text{for } E = 0 \text{ and } \theta_2 = 1 \rightarrow \ 6. \end{cases}
\end{aligned}$$

Note: The conditions $V_m^{A\beta} = 0.47 \in (0,1)$ and $V_m^{\text{retic}} = 0.93 \in (0,1)$ are fulfilled in our data. Furthermore, $k_d = 46400$ and $k_e = 1198.8$ are by far the biggest parameters. (table 6.4)
The following two figures (figure 6.5 and 6.6) illustrate our considerations:

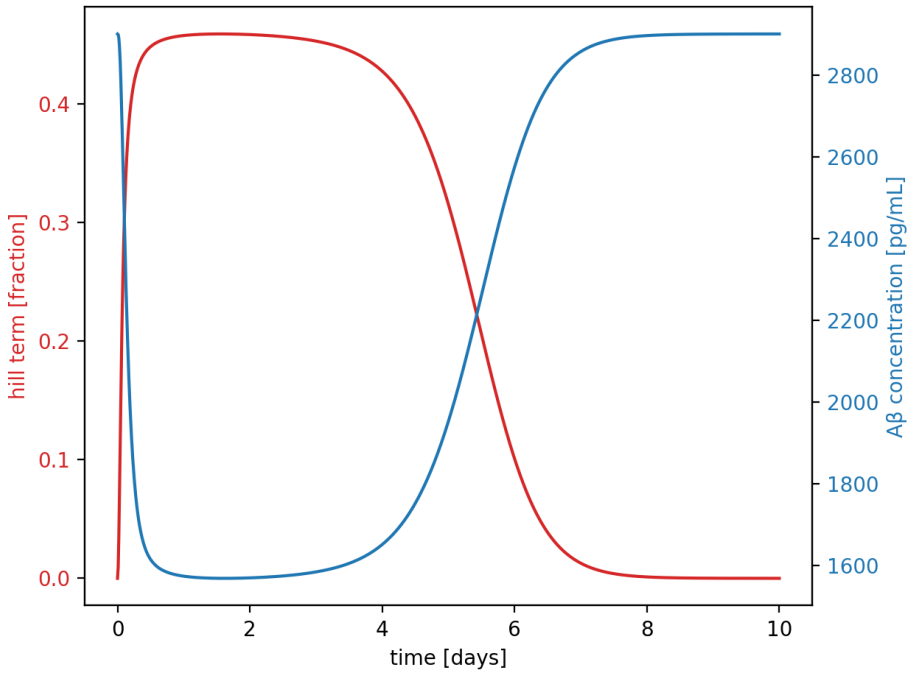


Figure 6.5: $h_1(H)$ (6.7) and amyloid-β concentration $D(t)$ for the parameter and initial values in the tables 6.4 and 6.5

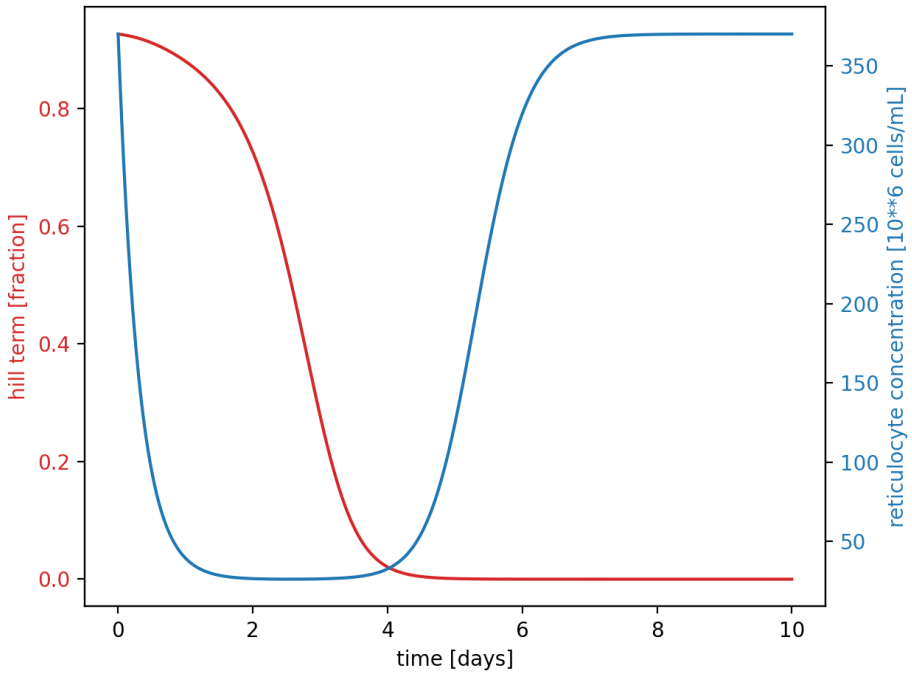


Figure 6.6: $h_2(A)$ (6.8) and reticulocyte concentration $E(t)$ for the parameter and initial values in the tables 6.4 and 6.5

6.3 The effect of the affinity parameters

6.3.1 Introduction of the five antibodies

In the article by Gadkar et al. 2016 [10] five bispecific antibodies were tested. The affinity parameters of those are shown in the table 6.2 below:

antibody	K_m^{TfR}	K_m^{retic}	K_m^{retic} approx with $p_1(x)$ (6.13)	unit
A	66.6	5.32	92.7	μg/mL
D	406.3	52.37	103.4	μg/mL
E	23031	756.2	813.9	μg/mL
Db	2500	250	169.2	μg/mL
Dc	10000	520	404.7	μg/mL

Table 6.2: Affinity parameters of the tested antibodies

In the next step we solve the equations (6.1) to (6.4) for $A(t)$ to $E(t)$ for the antibodies A, D, Db, Dc and E:

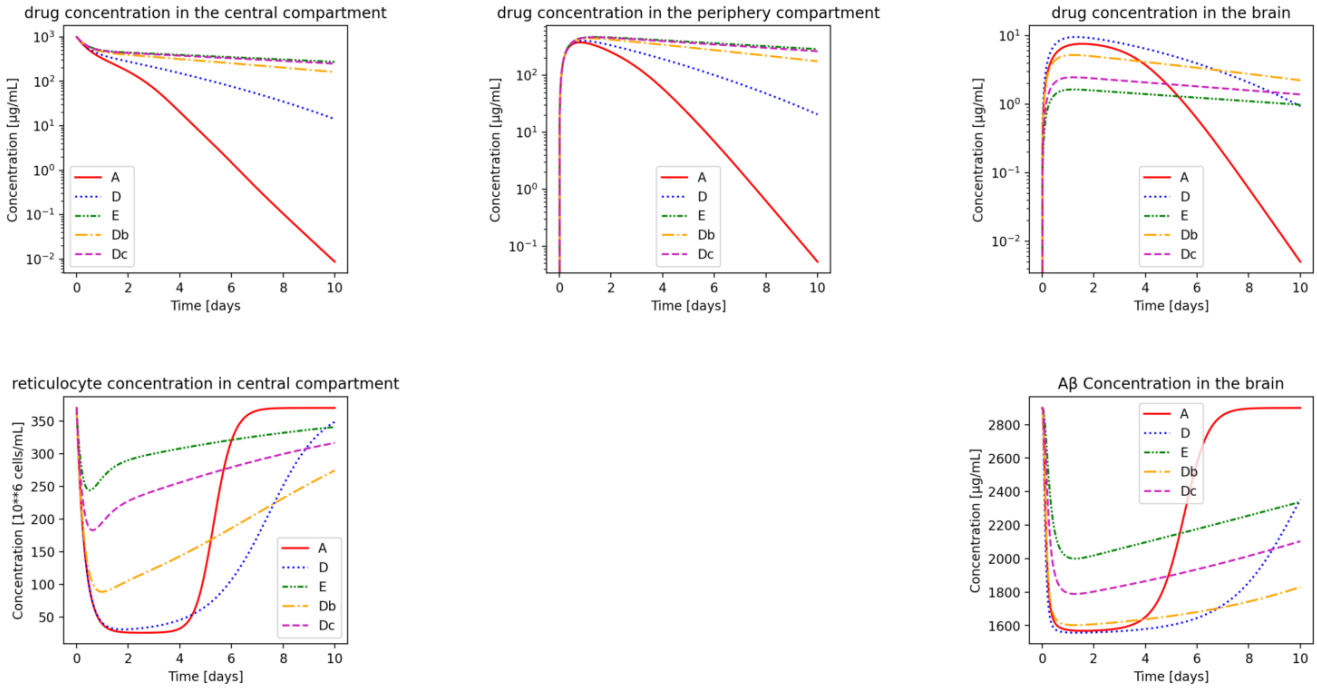


Figure 6.7: Solution of the model (6.1) to (6.4) with parameter values 6.4 and IVs 6.5, differ in transferrin and reticulocyte affinity 6.2

The solutions shown in figure 6.7 displays the same dynamics as shown in figure 6.4, but for five different antibodies with different affinity parameters. The properties of these antibodies

were provided in the article by Gadkar et al. 2016 [10].

6.3.2 Interpolating the dependencies of the TfR-affinity vs the reticulocyte-affinity

We are interested in the dependency of the affinity parameters (table 6.2). We use numerical interpolation to find a polynomial of second degree, which gives us insights into the dependency.

$$p_2(x) = ax^2 + bx + c \text{ with } a = -1.51 \cdot 10^{-6},\ b = 6.60 \cdot 10^{-2},\ c = 34 \tag{6.12}$$

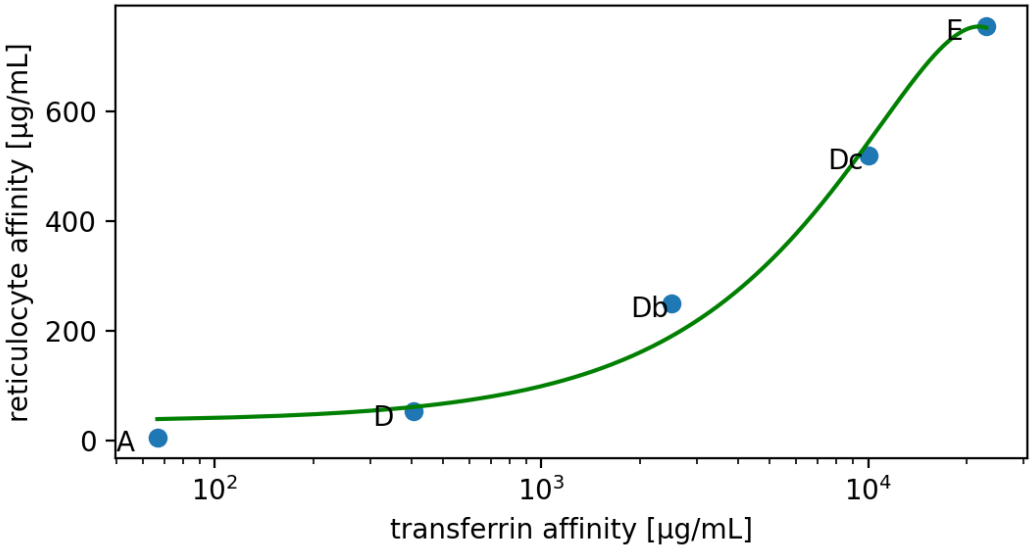


Figure 6.8: The affinity parameters of the antibodies in the table 6.2 and the quadratic polynomial interpolation function

The problem with the function $p_2(t)$ is that, although it fits really well, we run into the issue of $a < 0$. From $\frac{dp_2}{dt} = 0 \Leftrightarrow -2at = b \Leftrightarrow t = \frac{b}{-2a} \approx 21900$ we get that the for transferrin affinities greater than approximately $21900\frac{\mu g}{mL}$ the transferrin and the reticulocyte affinities are negatively correlated. There is no evidence for this from the data. Since we are mainly interested in the correlation of these affinities, this approach is of little value. Adding more constraints (like defining the value of the derivative at one point) seems due to the limited data not very reasonable. Instead, we reduce the complexity of the interpolation and use a linear model:

$$p_1(x) = mx + c \text{ with } m = 3.14 \cdot 10^{-2},\ c = 90.7 \tag{6.13}$$

The derivative is constant $\frac{dp_c}{dx} = m$, which makes it easy to work with.
To evaluate the quality of the linear polynomial approximation (6.13), we compare figure 6.7 to

the solution with reticulocyte affinities calculated with $p_1(x)$:

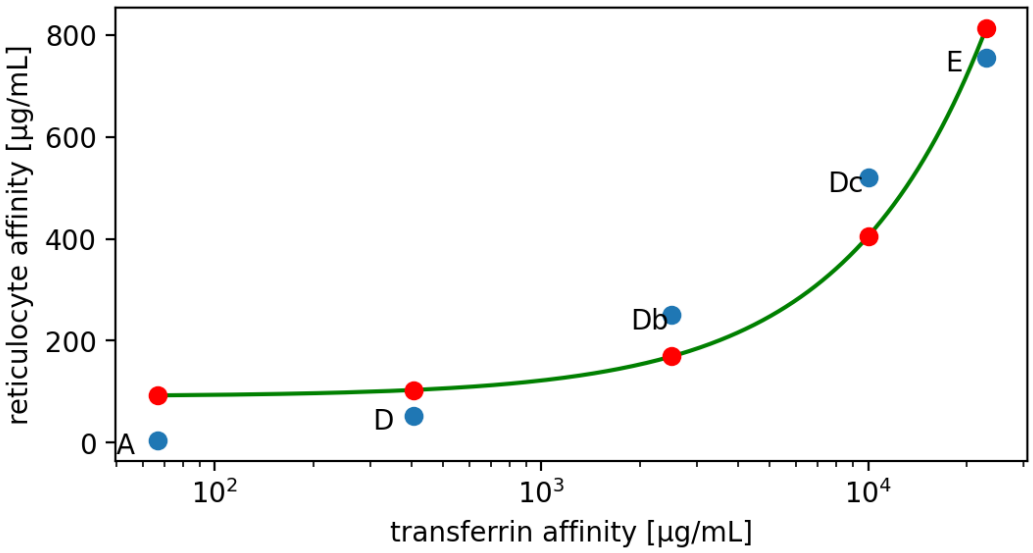


Figure 6.9: The affinity parameters of the antibodies in the table 6.2 (blue), the linear interpolation (green) and the interpolated points (red), the x-axis is logarithmic making the linear function curved

To evaluate the quality of the linear fitting we define the following metric to measure the "Safety Area":

Definition 5 (Safety Area)

$$A_t\left(K_m^{TfR}, K_m^{TfR}\right) = \int_0^t \left(E_0 - E_{\left(K_m^{TfR}, K_m^{TfR}\right)}(t)\right) dt$$

E_0 is the initial reticulocyte concentration (table 6.5)

If the safety area is positive, the reticulocyte concentration drops below the initial concentration within the time frame. We already know that this is a side effect of the drug. This was the original assumption leading to the model.

Lemma 2 (Positivity of the Safety Area)
Let:
$E_0 > E(\tilde{t}) \;\forall\; \tilde{t} \in C \subset [0,t]$
Than is the Safety Area positive.

Proof:
$A_t > 0 \overset{\text{Def. 5, E(t) continuous, } t>0}{\iff} E_0 > E(\tilde{t}) \;\forall\; \tilde{t} \in C \subset [0,t]$

■

To find constraints of the linear fitting approach (6.13) we compare the safety areas calculated with the given affinity parameters (table 6.2) and these with approximated K_m^{retic} (figure 6.9). The next figure shows the safety are normalized with 100% being the initial reticulocyte concentration multiplied with the time span, which is set to 10 days.

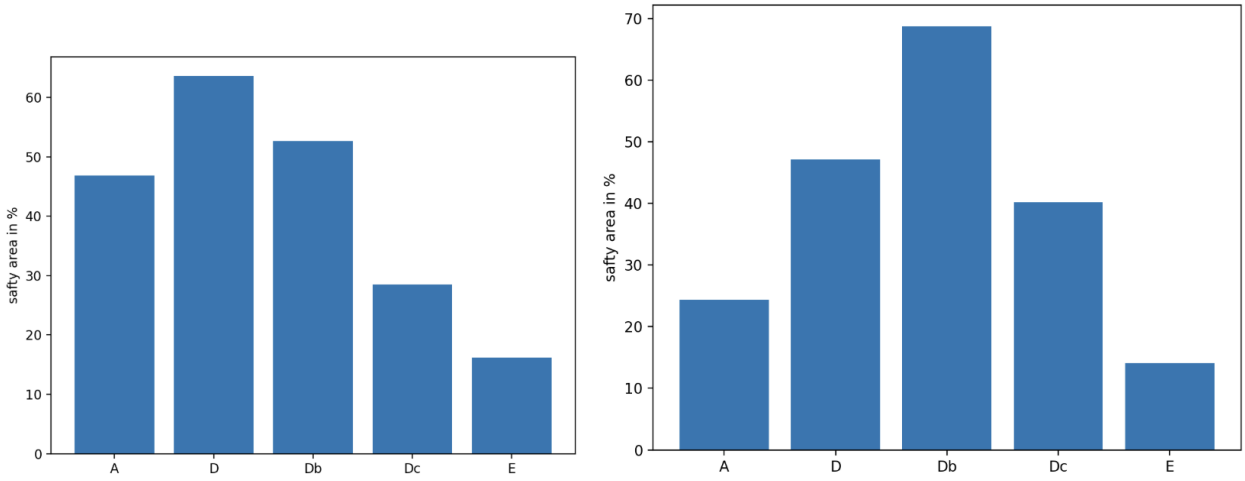


Figure 6.10: Safety area for the five antibodies: given K_m^{retic} left, approximated K_m^{retic} right

The result is disappointing: Not only the values differ, but they also differ in comparison. While in the left diagram antibody D gets the best result (largest safety area), in the right diagram antibody Db gets the best result. This means, that the approximation $p_1(x)$ (6.13) does not give us valuable insights about: Which is the antibody with the least impact on the reticulocyte concentration? We need to come up with a new approximation with the following conditions: The function has always a positive derivative like $p_1(x)$ (6.13), but is a more accurate approximation. A different approach is interpolation with a linear spline:

$$s_3(x) = \sum_{i=0}^{3} (m_i x + c_i)\, \mathbb{1}_{x\in[x_i,x_{i+1}]} \text{ with } m_i = \frac{y_i - y_{i+1}}{x_0 - x_{x_{i+1}}} \text{ and } c_i = y_i - m_i x_i \text{ for the data } (x_0, y_0)...(x_4, y_4) \tag{6.14}$$

	m_i	c_i
0	0.139	-3.904
1	0.0944	14.02
2	0.036	160
3	0.0181	338.7

Table 6.3: the values for m_i and c_i for the linear spline $s_3(x)$ (6.14)

We notice that there is no extreme variation within the m_i values (especially no negative values). Comparing the m_i values from table 6.3 to the calculated $m = 3.14 \cdot 10^{-2}$ in the linear polynomial (6.13), there is no sign of overfitting. Actually since $m_3 = 0.0181 < 0.0314$ and the approximated K_m^{retic} for the antibody E (with $p_1(x)$ (6.13)) is already higher than the affinity from the data for E (compare figure 6.8), we get with the spline interpolation better results for higher affinities ($p_1(x)$ overestimates the reticulocyte affinity for high transferrin affinities). Despite being not continuously differentiable like a polynomial, the linear spline $s_3(x)$ is piece wise differentiable. This is absolutely sufficient for our purpose.

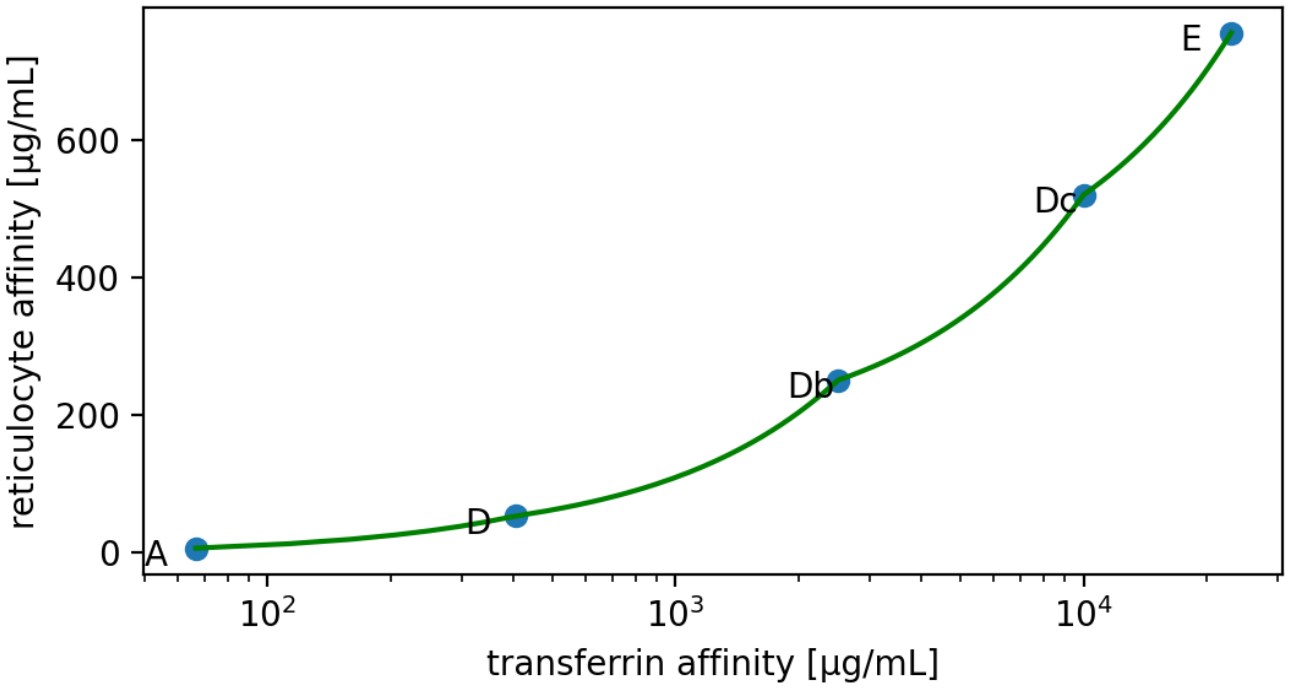


Figure 6.11: Interpolation of K_m^{retic} with linear splines, the x-axis is logarithmic making the linear splines curved

The spline interpolation perfectly fits the data points (by design), so there is no comparison of the safety areas necessary.

6.3.3 Sensitivity of the model to the affinity parameters

Now since we found a good interpolation for the affinities, we can plot the calculated concentrations $A(\tilde{t})$, $B(\tilde{t})$, $H(\tilde{t})$, $D(\tilde{t})$ and $E(\tilde{t})$ at a certain time $\tilde{t} = 2$ [days] as functions of the transferrin affinity K_m^{TfR} and reticulocyte affinity K_m^{retic}. $K_m^{\text{retic}} \sim s_3(K_m^{\text{TfR}})$ (6.14) is approximated with the linear spline function. From the model ((6.1) - (6.5)) and from the figure 6.2 we recognize, that while K_m^{TfR} has an impact on all equations, K_m^{retic} appears only in $\dot{E}$ (6.5) and none of the other terms depend on E (concentration of reticulocyte). This means, that approximated value of K_m^{retic} has only an impact on one of the five concentrations, as we will see in the following plot. The plot for the reticulocyte concentration will show the effect of K_m^{TfR} and the effect of K_m^{TfR} together with the approximated K_m^{retic} as two graphs. For the constant reticulocyte affinities $K_m^{\text{retic}} = 520 \frac{\mu g}{mL}$ is chosen. This is the value for antibody "Dc".

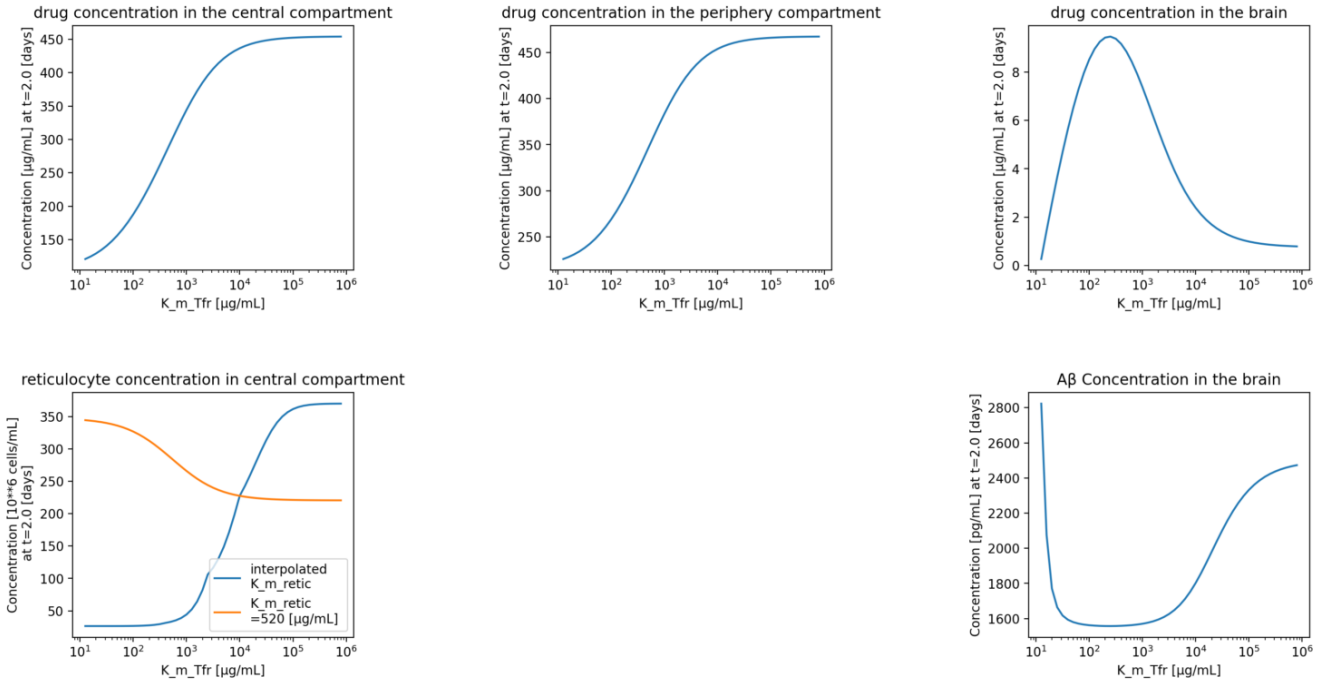


Figure 6.12: Effect of the affinities on the concentrations on day 2

Now we compare the results shown in figure 6.12 with our previous results (figure 6.7) and our expectations:

A(t) and P(t): drug concentration in central and periphery compartment

Lower K_m^{TfR} leads to faster decrease of the drug in the central and the periphery compartment. Figure 6.7 shows that the antibody "A" with the lowest affinity parameters has the lowest drug concentration of all antibodies for the whole timescale. Furthermore, none of the five graphs intersect in figure 6.7. This means, that we have a similar behavior for other time points.

H(t): drug concentration in the brain

Figure 6.7 shows that the antibodies with lower transferrin affinity have a steeper increase and steeper decrease than those with higher affinities. Furthermore, antibody "E" with $K_m^{\mathrm{TfR}} = 406.3 \frac{\mu g}{mL}$ (table 6.2) has the highest drug concentration during the first days. In figure 6.12 we see the peak of drug concentration in brain at parameters in the range between 10^2 and 10^3. The fact that the graphs are intersecting in figure 6.7 means that for later time points the peak will be at a different value.

D(t): amyloid-β concentration in the brain

Since the drug is designed to reduce the amyloid-β concentration in the brain, we see contrary behavior comparing the amyloid-β and the drug concentration in brain in both figures. This means if the drug concentration drops the amyloid-β concentration increases and vice versa.

E(t): reticulocyte concentration in central compartment

This is the critical part of the model. As discussed we want the side effect of the drug on the reticulocyte concentration to be as low as possible. Here both affinities K_m^{retic} and K_m^{TfR} have an impact. We see the opposite behavior (in figure 6.12), if we approximate K_m^{retic}: The reticulocyte concentration would decease with rising K_m^{TfR} on day 2, if we only consider K_m^{TfR}, but increases if we take also K_m^{retic} into account. Figure 6.7 verifies this behavior at day 2, but suggest since the graphs intersect, that we should investigate the dynamics at a different time point.

The behavior on day 5

At a later time point we see that the plots for drug concentration and amyloid-β concentration correspond to those on day 2. The only difference is, that everything is shifted to the left. The reticulocyte concentration shows the expected more complex behavior, which makes our results interesting for optimization.

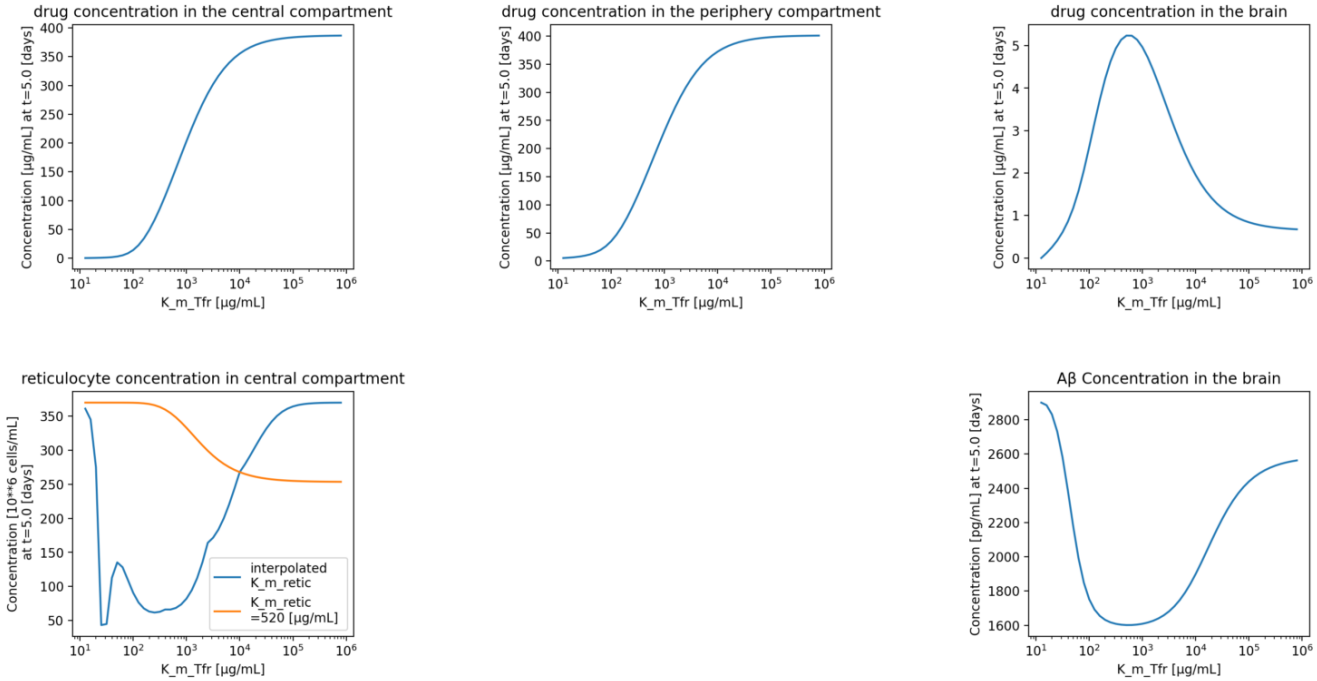


Figure 6.13: Effect of the affinities on the concentrations on day 5

6.4 Data

6.4.1 Parameter values

	abbreviation	describtion	value	unit
0	C_L	drug degradation in central comp.	4.73	mL/day/kg
1	C_{LD}	exchange of drug between central and perifery periphery comp.	71.01	mL/day/kg
2	V_1	volume central comp.	51.95	mL/kg
3	V_2	volume periphery comp.	48.25	mL/kg
4	V_m^{BACE}	BRACE volume	241.1	μg/mL
5	K_m^{BACE}	BRACE affinity	1.3	μg/mL
6	V_m^{TfR}	TfR volume	16061	μg/day/kg
7	k_h	nonspecific uptake from central to periphery compartment	0.00264	1/day
8	k_d	synthesis of amyloid beta	46400	pg/nL/day
9	k_e	formation of reticulocyte	1198.8	million cells/mL/day
10	α	alpha	21.5	no unit
11	β	beta	-23.8	no unit
12	l_h	drug degradation in brain	1.64	1/day
13	l_d	amyloid beta degradation in brain	16	1/day
14	l_e	reticulocyte degradation in central comp.	3.24	1/day
15	IC50	half maximal inhibitory concentration	1.18	μg/mL
16	n	hill coefficient for H	2	no unit
17	n_1	hill coefficient for A	2.4	no unit
18	$V_m^{A\beta}$	amyloid beta fraction	0.47	fraction
19	V_m^{retic}	reticulocyte fraction	0.93	fraction

Table 6.4: parameters

6.4.2 Initial values

abbreviation	concentration	value	unit
A	central (drug)	1000	μg/mL
P	periphery (drug)	0	μg/mL
H	brain (drug)	0	μg/mL
D	amyloid beta in brain	2900	pg/mL
E	reticulocyte in central comp.	370	10^6 cells/mL

Table 6.5: initial values

7 Modeling the Uptake of TfRMAb through the BBB in Detail

This chapter is based on the article by W. Pardridge and T. Chou, 2021 [13]. The ODEs (7.1) - (7.11) and following and parameter values come from [13].

7.1 Receptor mediated uptake of monoclonal antibodies

In this chapter we model the receptor medicated uptake of monoclonal antibodies (MAb) targeting the transferrin receptor (TfR). We have already discussed in chapter 6, that the ability of a MAb to bind to the TfR induces receptor medicated transport (RMT) across the blood brain barrier (BBB). (figure 6.1) The mass action kinetic model will be given as ODE, which can be solved numerically and plotted. The model from the article [13] will be simplified, and the simplification will be justified.

7.2 Model

The model displays the uptake of the drug (the TfRMAb) from the blood plasma to the brain. In contrast to the model in chapter 6 this model does not describe the effect of the drug in the brain and in the plasma, but describes in detail what happens in the BBB. This means that in this chapter we get detailed insights, about what happens between the compartments "drug (central)" and "drug (brain)" in the model from chapter 6. The drugs in the two chapters differ and the values in this chapter come from Rhesus monkeys instead of mice. This means the values will differ a lot, but the dynamics are comparable.

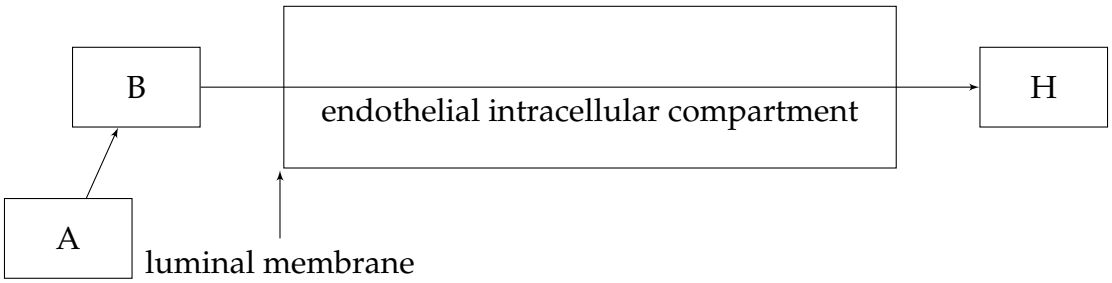


Figure 7.1: The uptake of the TfRMAb through the BBB, which consists of the luminal membrane and the endothelial intracellular compartment. $A(t)$ is the input function, and $B(t)$ and $H(t)$ are variables of the model, which is described in the next figure (7.2) more detailed.

The figure 7.1 shows how the drug has to go through the luminal membrane of the BBB and intracellular compartment of the BBB before reaching in the brain. The terms $A(t)$, $B(t)$ and $H(t)$ describe the concentration of the drug (TfRMAb) in the arterial plasma ($A(t)$), capillary plasma ($B(t)$) and brain extracellular space ($H(t)$):

Definition 6 (Concentration of TfRMAb in Arterial Plasma)
The concentration of TfRMAb in arterial plasma is defined by the function $A(t)$ (input function):

1. *one initial dose: $A(t) := A_0 e^{-\alpha t}$ for $t \geq 0$, $A_0 > 0$, $\alpha > 0$ (exponential decline of TfRMAb concentration)*
2. *constant dose: $A(t) = A_0$ for $A_0 > 0$*
3. *periodic dose: $A(t) := A_0 e^{-\alpha(\tilde{t}+t) \text{ (modulo) } p}$ with the time shift $\tilde{t} \geq 0$ and period $p > \tilde{t}$.*

The reason for the $\tilde{t}$ shift is, that we want to start for initial values $\tilde{t}$ minutes after drug injection. This means that for $t = 0$ the drug dose in the arterial plasma is already at a lower level than the initial dose. This dose is injected again at $t = p - \tilde{t}$.
The drug is injected intravenously. So the term $A(t)$ can be controlled directly.

Definition 7 (Concentatrion of TfRMAb in the other compartments)

- *$B(t)$: transferrin receptor monoclonal antibodies (TfRMAb) in capillary plasma*
- *$H(t)$: transferrin receptor monoclonal antibodies (TfRMAb) in brain extracellular space*

The multistep process in the brain capillary endothelium is shown in the model (figure 7.2). All boxes represent the concentration of one compartment, which are defined as ODE. Since $A(t)$ is given explicitly as input, $A(t)$ is not symbolized by a box.

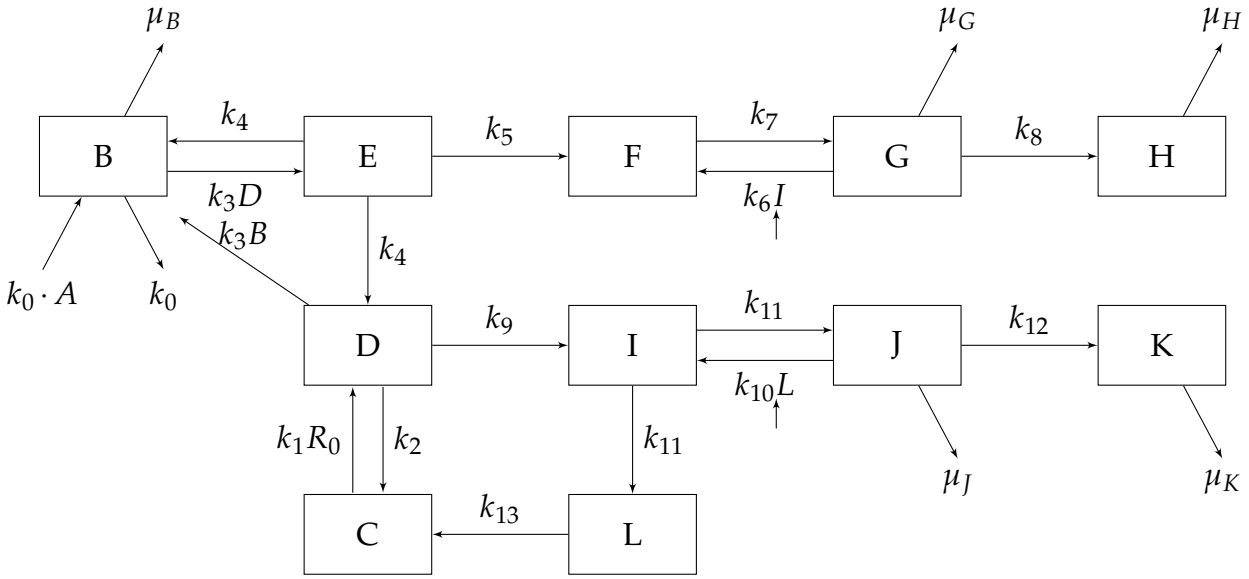


Figure 7.2: Model from [13] with all terms and without any simplifications. The concentration terms, which are the variables of the model, are replaced by the blocks, and the arrows represent the dynamics.

We go through to terms of the compartments of the BBB step by step starting with the luminal membrane:

Definition 8 (Concentrations at the Luminal Membrane)

- $C(t)$: *unoccupied transferrin receptor at luminal membrane*
- $D(t)$: *transferrin receptor-transferrin (TfR-Tf) complex at luminal membrane*
- $E(t)$: *transferrin receptor-transferrin-transferrin receptor monoclonal antibodies (TfR-Tf-TfRMAb) complex at luminal membrane*

Next we define the five concentrations in the endothelial intracellular compartment:

Definition 9 (Concentrations in the endothelial intracellular compartment)

- $L(t)$: *unoccupied transferrin receptor in the endothelial intracellular compartment*
- $I(t)$: *transferrin receptor-transferrin (TfR-Tf) complex in the endothelial intracellular compartment*
- $F(t)$: *TfR-Tf-TfRMAb complex in the endothelial intracellular compartment*
- $J(t)$: *unbounded transferrin in the endothelial intracellular compartment*
- $G(t)$: *unbounded TfRMAb in the endothelial intracellular compartment*

The last variable in the system is the transferrin, that is transport to the brain. The concentration of TfRMAb in brain extracellular space ($H(t)$) is already defined.

Definition 10 (Concentatrion transferrin in the brain)

- $K(t)$: *transferrin in brain extracellular space*

To summarize these definitions: The first row of the figure B, E, F, G, H represents the bounded and unbounded drug (TfRMAb) in the different compartments. The second row D, I, J, K represents the bounded and unbounded transferrin. The third row C, L represents the unoccupied transferrin receptor (TfR). The columns represent the steps of the drug uptake through the BBB: blood plasma, luminal membrane, endothelial intracellular compartment and brain (left to right).
After introducing all variables of the model (figure 7.2), we can describe their dependencies and interactions with ODEs:

$$\dot{B} = k_0 (A - B) - k_3 DB - \mu_B B + k_4 E \tag{7.1}$$
$$\dot{C} = k_{13} L - k_1 C R_0 + k_2 D \tag{7.2}$$
$$\dot{D} = k_1 C R_0 - k_2 D - k_9 D - k_3 DB + k_4 E \tag{7.3}$$
$$\dot{E} = k_3 DB - k_4 E - k_5 E \tag{7.4}$$
$$\dot{F} = k_5 E + k_6 IG - k_7 F \tag{7.5}$$
$$\dot{G} = k_7 F - k_6 IG - k_8 G - \mu_G G \tag{7.6}$$
$$\dot{H} = k_8 G - \mu_H H \tag{7.7}$$
$$\dot{I} = k_9 D + k_{10} LJ - k_{11} I + k_7 F - k_6 IG \tag{7.8}$$
$$\dot{J} = k_{11} I - k_{10} LJ - k_{12} J - \mu_J J \tag{7.9}$$
$$\dot{K} = k_{12} J - \mu_K K \tag{7.10}$$
$$\dot{L} = k_{11} I - k_{13} L - k_{10} LJ \tag{7.11}$$

7.2.1 One initial dose of TfRMAb

For this section we assume the input function $A(t)$ (Definition 6) to be exponentially declining with A_0 = 10.34 nM (concentration of TfRMAb in arterial plasma at t=0) and α = $0.0055\frac{1}{min}$ (removal of TfRMAb from plasma by peripheral tissues). These values are used to find the numerical solutions for the figures 7.3 and 7.5, and to find the values for $\bar{C}$ in Lemma 3 and Lemma 4 by studying the long term behavior of the numerical solution. Despite that the proofs and the statements of Lemma 3 and Lemma 4 are independent of the chosen A_0 and α values (compare Theorem 7 with variable C=X).
The next figure 7.3 shows the numerical solutions of the ODEs with the parameter values from table 7.5 and table 7.6 and the initial values from table 7.7. The initial values describe the concentrations of the compartments after injecting an initial dose of 0.2 $\frac{mg}{kg}$ to a Rhesus monkey after 2 hours or 120 minutes. This means the maximal plasma concentration of TfRMAb (20nM) is at time t = −120 minutes and at t=0 A_0 = 10.34 nM. We assume the total transferrin in plasma to be R_0 = 25000 nM and the total TfR in endothelial cell to be L_0 = 40 nM at t = −120 minutes.

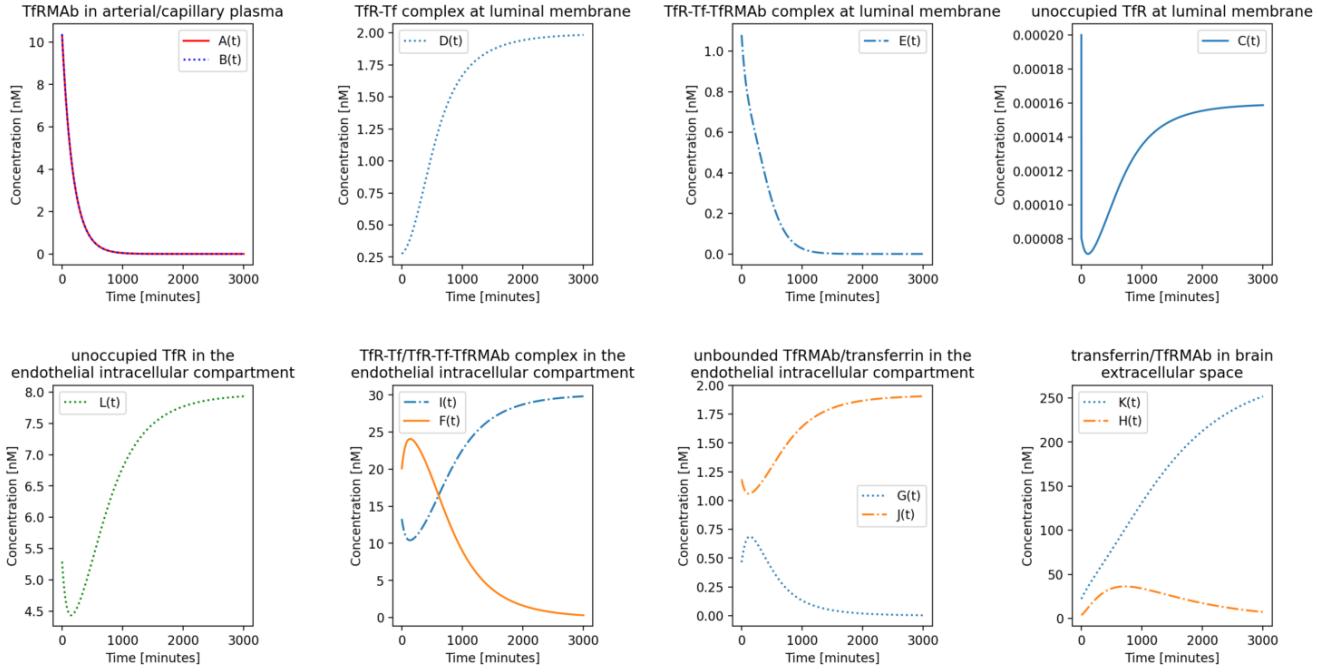


Figure 7.3: Numerical solution of the model (figure 7.2) with all compartments and one initial dose (parameter values from table 7.5, table 7.6, IVs from table 7.7, $R_0 = 25000$, $A_0 = 10.34$ and $\alpha = 0.0055\frac{1}{min}$)

For $A(t)$ declining explicitly (Definition 6. 1.) the terms $B(t)$, $E(t)$, $F(t)$, $G(t)$ and $H(t)$ tend to 0. These are the terms, which describe the bounded and unbounded TfRMAb. Biologically this is necessary for the model, since without further drug injection the TfRMAb will not stay forever in the body. The following theorem gives us one steady state for with TfRMAb concentration in all compartments being 0 and the dependencies of the other terms.

Theorem 7

Let:

$\bar{B} = 0$, $\bar{C} = X$, $\bar{D} = \lambda_1 X$, $\bar{E} = 0$, $\bar{F} = 0$, $\bar{G} = 0$, $\bar{H} = 0$, $\bar{I} = \frac{1}{k_{11}}\left(k_9\lambda_1 X + k_{10}\lambda_2\lambda_3 X^2\right)$, $\bar{J} = \lambda_3 X$, $\bar{K} = \lambda_4 X$, $\bar{L} = \lambda_2 X$ *and with:*

$\lambda_1 = \frac{k_1 R_0}{k_2+k_9}$, $\lambda_2 = \frac{1}{k_{13}}\left(k_1 R_0 - k_2\lambda_1\right)$, $\lambda_3 = \frac{k_9}{k_{12}+\mu_J}\lambda_1$, $\lambda_4 = \frac{k_{12}\lambda_3}{\mu_K}$ *and* $X \geq 0$.

Then:

The system displayed in figure 7.2 and described with the ODEs ((7.1)-(7.11)) has the following steady state for $A = 0$*:*

$(B, C, D, E, F, G, H, I, J, K, L) = \left(\bar{B}, \bar{C}, \bar{D}, \bar{E}, \bar{F}, \bar{G}, \bar{H}, \bar{I}, \bar{J}, \bar{K}, \bar{L}\right)$

Proof:

$$\dot{B} \overset{(7.1)}{=} k_0(\underbrace{\bar{A}}_{=0} - \underbrace{\bar{B}}_{=0}) - k_3\bar{D}\underbrace{\bar{B}}_{=0} - \mu_B\underbrace{\bar{B}}_{=0} + k_4\underbrace{\bar{E}}_{=0} = 0$$

$$\begin{aligned}\dot{C} &\overset{(7.2)}{=} k_{13}\bar{L} - k_1\bar{C}R_0 + k_2\bar{D} \\ &= k_{13}\lambda_2 X - k_1 X R_0 + k_2\lambda_1 X \\ &= \frac{k_{13}}{k_{13}}\left(k_1 R_0 - k_2\lambda_1\right) X - k_1 X R_0 + k_2\lambda_1 X = 0\end{aligned}$$

$$\begin{aligned}\dot{D} &\overset{(7.3)}{=} k_1\bar{C}R_0 - k_2\bar{D} - k_9\bar{D} - k_3\bar{D}\underbrace{\bar{B}}_{=0} + k_4\underbrace{\bar{E}}_{=0} \\ &= k_1\bar{C}R_0 - k_2\bar{D} - k_9\bar{D} - k_3\bar{D} \\ &= k_1 X R_0 - k_2\lambda_1 X - k_9\lambda_1 X \\ &= k_1 X R_0 - k_2\frac{k_1 R_0}{k_2 + k_9} X - k_9\frac{k_1 R_0}{k_2 + k_9} X \\ &= k_1 R_0 X - \frac{k_2 + k_9}{k_2 + k_9} k_1 R_0 X = 0\end{aligned}$$

$$\dot{E} \overset{(7.4)}{=} k_3 D\underbrace{\bar{B}}_{=0} - k_4\underbrace{\bar{E}}_{=0} - k_5\underbrace{\bar{E}}_{=0} = 0$$

$$\dot{F} \overset{(7.5)}{=} k_5\underbrace{\bar{E}}_{=0} + k_6 I\underbrace{\bar{G}}_{=0} - k_7\underbrace{\bar{F}}_{=0} = 0$$

$$\dot{G} \overset{(7.6)}{=} k_7\underbrace{\bar{F}}_{=0} - k_6 I\underbrace{\bar{G}}_{=0} - k_8\underbrace{\bar{G}}_{=0} = 0$$

$$\dot{H} \overset{(7.7)}{=} k_8\underbrace{\bar{G}}_{=0} - \mu_H\underbrace{\bar{H}}_{=0} = 0$$

$$\begin{aligned}\dot{I} &\overset{(7.8)}{=} k_9\bar{D} + k_{10}\bar{L}\bar{J} - k_{11}\bar{I} + k_7\underbrace{\bar{F}}_{=0} - k_6 I\underbrace{\bar{G}}_{=0} \\ &= k_9\bar{D} + k_{10}\bar{L}\bar{J} - k_{11}\bar{I} \\ &= k_9\lambda_1 X + k_{10}\lambda_2 X\lambda_3 X - k_{11}\frac{1}{k_{11}}\left(k_9\lambda_1 X + k_{10}\lambda_2\lambda_3 X^2\right) = 0\end{aligned}$$

$$\begin{aligned}\dot{J} &\overset{(7.9)}{=} k_{11}\bar{I} - k_{10}\bar{L}\bar{J} - k_{12}J - \mu_J\bar{J} \\ &= \frac{k_{11}}{k_{11}}\left(k_9\lambda_1 X + k_{10}\lambda_2\lambda_3 X^2\right) - k_{10}\lambda_2 X\lambda_3 X - k_{12}\lambda_3 X - \mu_J\lambda_3 X \\ &= k_9\lambda_1 X + k_{10}\lambda_2\lambda_3 X^2 - k_{10}\lambda_2\lambda_3 X^2 - \left(k_{12} + \mu_J\right)\lambda_3 X \\ &= k_9\lambda_1 X - \left(k_{12} + \mu_J\right)\frac{k_9}{k_{12} + \mu_J}\lambda_1 X = 0\end{aligned}$$

$$\begin{aligned}
\dot{K} &\overset{(7.10)}{=} k_{12}\bar{J} - \mu_K \bar{K} \\
&= k_{12}\lambda_3 X - \mu_K \lambda_4 X \\
&= k_{12}\lambda_3 X - \mu_K \frac{k_{12}\lambda_3}{\mu_K} X \\
&= k_{12}\lambda_3 X - k_{12}\lambda_3 X = 0
\end{aligned}$$

$$\begin{aligned}
\dot{L} &\overset{(7.11)}{=} k_{11}\bar{I} - k_{13}\bar{L} - k_{10}\bar{L}\bar{J} \\
&= \frac{k_{11}}{k_{11}}\left(k_9\lambda_1 X + k_{10}\lambda_2\lambda_3 X^2\right) - \frac{k_{13}}{k_{13}}\left(k_1 R_0 - k_2\lambda_1\right) X - k_{10}\lambda_2\lambda_3 X^2 \\
&= k_9\lambda_1 X - k_1 R_0 X + k_2\lambda_1 X \\
&= \left(\frac{k_9 k_1 R_0}{k_2 + k_9} - k_1 R_0 + \frac{k_2 k_1 R_0}{k_2 + k_9}\right) X \\
&= \left(k_9 k_1 R_0 - k_1 R_0 k_2 + k_1 R_0 k_9 + k_2 k_1 R_0\right) \frac{X}{k_2 + k_9} = 0
\end{aligned}$$

We get $\left(\dot{B}, \dot{C}, \dot{D}, \dot{E}, \dot{F}, \dot{G}, \dot{H}, \dot{I}, \dot{J}, \dot{K}, \dot{L}\right) = (0,0,0,0,0,0,0,0,0,0,0)$ for (7.1) - (7.11), which concludes the proof.

■

The following values were found by computing the limit of the numerical solutions. The calculated derivatives were all close to 0 or exactly 0, but we will verify this in the following Lemma 3 and in its proof:

Lemma 3
The system displayed in figure 7.2 and described with the ODEs ((7.1)-(7.11)) has the following steady state for $A = 0$ *and the given parameter values from table 7.5, table 7.6 and* $R_0 = 25000$*:*
$\bar{B} = 0,\ \bar{C} = 1.59 \cdot 10^{-4},\ \bar{D} = 1.99,\ \bar{E} = 0,\ \bar{F} = 0,\ \bar{G} = 0,\ \bar{H} = 0,\ \bar{I} = 30,\ \bar{J} = 1.91,\ \bar{K} = 278,\ \bar{L} = 7.95$
These values are rounded.

Proof using Theorem 7:

$$\lambda_1 = \frac{k_1 R_0}{k_2 + k_9} = \frac{0.1 \cdot 25000}{0.06 + 0.14} = 12500$$
$$\lambda_2 = \frac{1}{k_{13}} (k_1 R_0 - k_2 \lambda_1) = \frac{1}{0.035} (0.1 \cdot 25000 - 0.06 \cdot 12500) = 50000$$
$$\lambda_3 = \frac{k_9}{k_{12} + \mu_J} \lambda_1 = \frac{0.14}{0.14 + 0.0058} \cdot 12500 = 12002.7$$
$$\lambda_4 = \frac{k_{12} \lambda_3}{\mu_K} = \frac{0.14 \cdot 12002.7}{0.00096} = 1750400$$
$$X = \bar{C} = 1.59 \cdot 10^{-4}$$
$$\bar{D} = \lambda_1 X = 12500 \cdot 1.59 \cdot 10^{-4} = 1.99$$
$$\bar{I} = \frac{1}{k_{11}} \left(k_9 \lambda_1 X + k_{10} \lambda_2 \lambda_3 X^2 \right)$$
$$= \frac{1}{0.06} \left(0.14 \cdot 12500 \cdot 1.59 \cdot 10^{-4} + 0.1 \cdot 50000 \cdot 12002.7 \cdot \left(1.59 \cdot 10^{-4} \right)^2 \right) = 30$$
$$\bar{K} = \lambda_4 X = 1750400 \cdot 1.59 \cdot 10^{-4} = 278$$
$$\bar{L} = \lambda_2 X = 50000 \cdot 1.59 \cdot 10^{-4} = 7.95$$

From Theorem 7 we get $\left(\dot{B}, \dot{C}, \dot{D}, \dot{E}, \dot{F}, \dot{G}, \dot{H}, \dot{I}, \dot{J}, \dot{K}, \dot{L} \right) = (0,0,0,0,0,0,0,0,0,0,0)$, which concludes the proof.

■

The following figure shows how the numerical solution converges against the calculated values from Lemma 3 $\bar{B} = 0$, $\bar{C} = 1.59 \cdot 10^{-4}$, $\bar{D} = 1.99$, $\bar{E} = 0$, $\bar{F} = 0$, $\bar{G} = 0$, $\bar{H} = 0$, $\bar{I} = 30$, $\bar{J} = 1.91$, $\bar{K} = 278$, $\bar{L} = 7.95$.

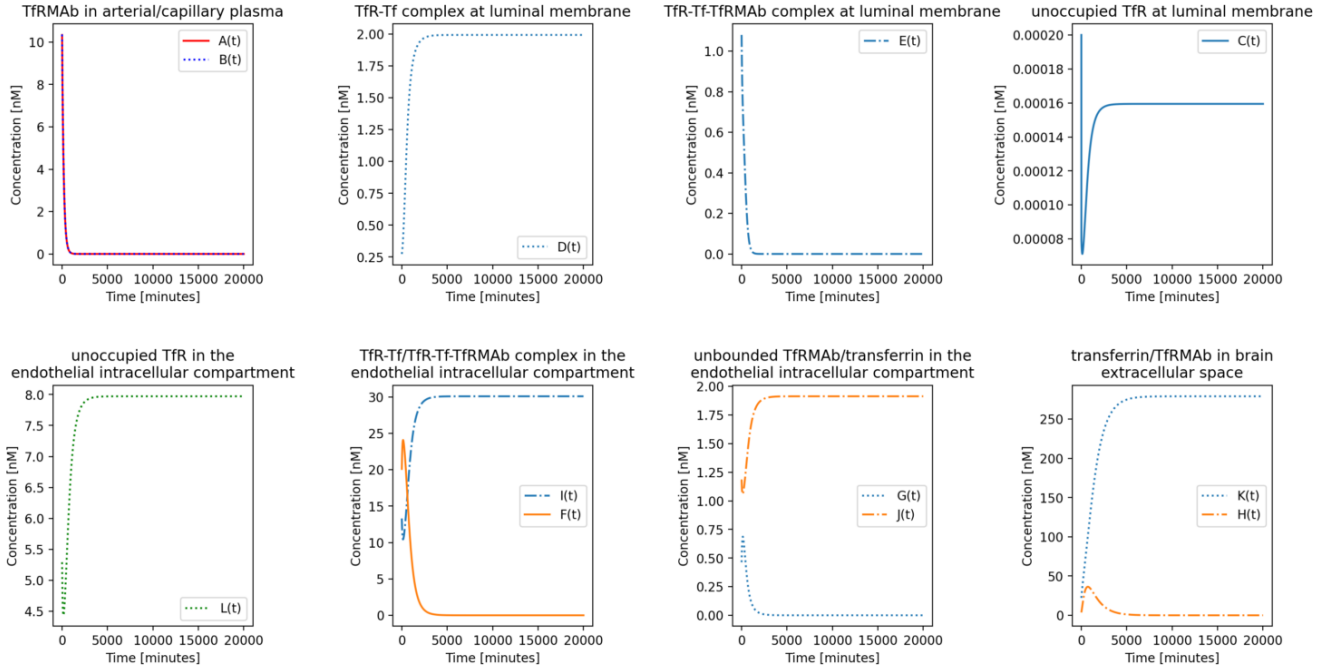


Figure 7.4: Numerical solution with the same parameter valuable as in figure 7.3 (parameter values from table 7.5, table 7.6, IVs from table 7.7, $R_0 = 25000$, $A_0 = 10.34$ and $\alpha = 0.0055\frac{1}{min}$), but for a longer time range to investigate the long term behavior and the steady states (compare Lemma 3)

7.2.2 Constant TfRMAb injection

On the other hand we are interested, what happens in the model, if we continuously administer the drug (Definition 6.2.) with $A_0 = 20$ nM. We have found those values by studying the long term behavior of the model 7.2, and prove that the values are steady states by calculating the derivatives:

Lemma 4

The system displayed in figure 7.2 and described with the ODEs ((7.1)-(7.11)) has the following steady state for $A = 20$ and the given parameter values from table 7.5, table 7.6 and $R_0 = 25000$:
$\bar{B} = 20$, $\bar{C} = 5.925 \cdot 10^{-5}$, $\bar{D} = 0.1198$, $\bar{E} = 26.18$, $\bar{F} = 26.17$, $\bar{G} = 0.8517$, $\bar{H} = 124.2$, $\bar{I} = 8.837$, $\bar{J} = 0.9667$, $\bar{K} = 140.98$, $\bar{L} = 4.027$
These values are rounded.

Proof in Appendix with more accurate values 9.4.

■

The following figure shows the solution of the model with the steady states also chosen to be the initial values, since we assume the drug is taken constantly over a long term period:

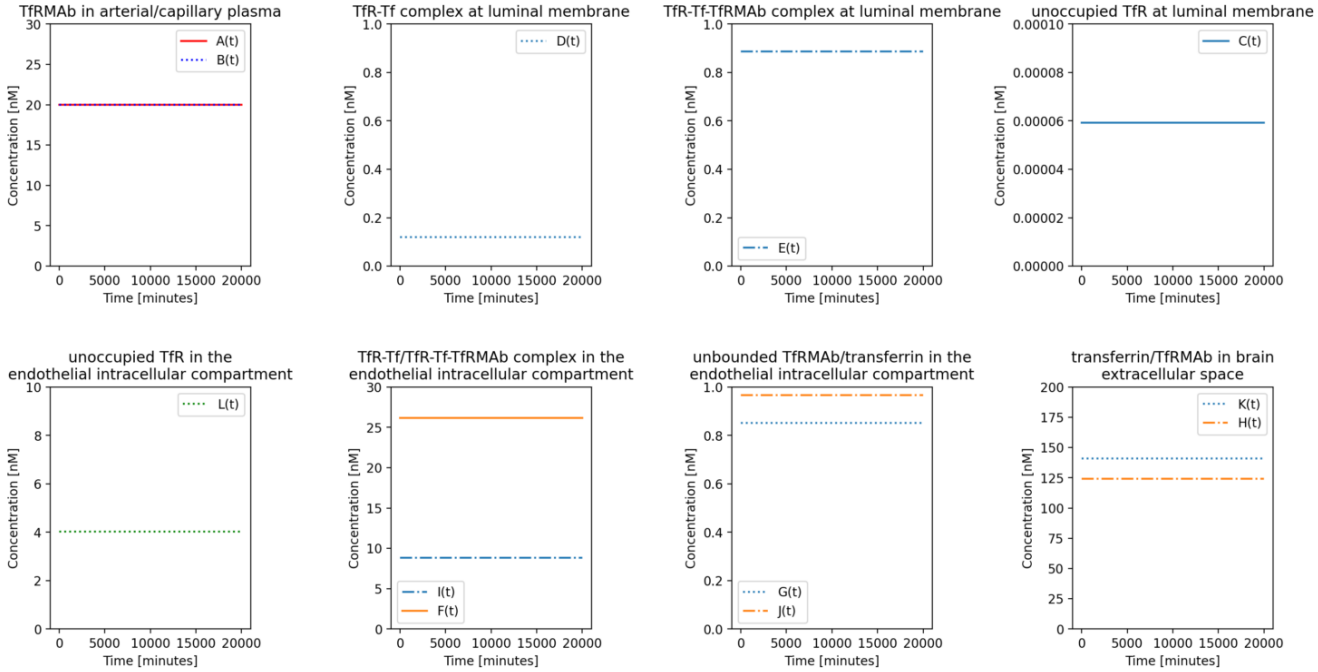


Figure 7.5: The steady states from Lemma 4 plotted for $A_0 = 20nM$ (constant) and the other given conditions of Lemma 4

7.2.3 Frequent TfRMAb injection

For this section we assume the input function $A(t)$ (Definition 6) to be exponentially declining over a period $p > \tilde{t} = 120$ min by the rate $\alpha = 0.0055\frac{1}{min}$ (removal of TfRMAb from plasma by peripheral tissues), and jumping (discontinuously) to the maximum concentration of TfRMAb in arterial plasma ($A_0 = 20$ nM) for the time points $t = np - \tilde{t}$ for $n \in \mathbb{N}$. (compare Definition 6.3.) $\tilde{t}$ is a time shift, which we add to the model since we also assumed in section 7.2.1, that the first dose was taken 120 min before $t = 0$. This means, that in the time span $t = 0$ to $t = p - \tilde{t}$ this model is identical to the model in section 7.2.1. Furthermore, we choose $R_0 = 25000$ as in section 7.2.1.

Now we analyze the models' behavior, if we inject the drug frequently with a period of $p = 720$ minutes or 12 hours. (Definition 6.3.) The following figure 7.6 shows this scenario for 2.5 days (3600 min):

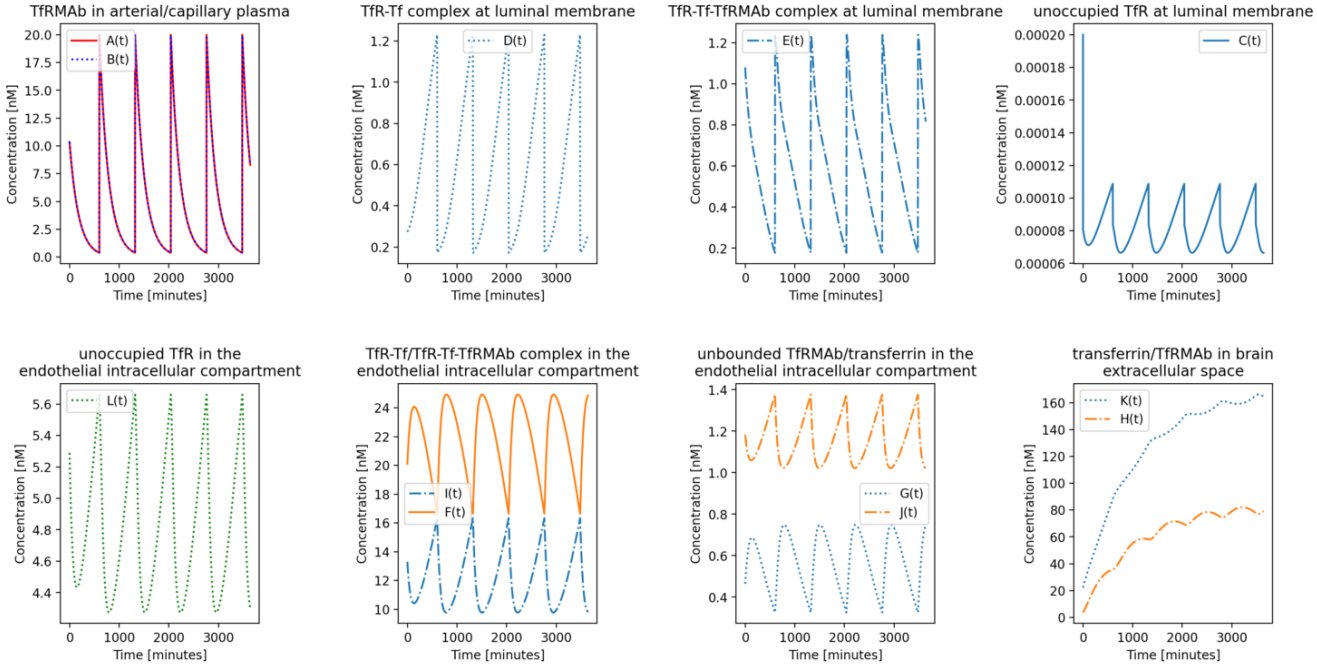


Figure 7.6: Dose 20nM is given every 12 hours (720 minutes)

As in the models before we assume that the drug is taken 120 minutes before $t = 0$. (Definition 6. 3. for $\tilde{t} = 120$ and IVs table 7.7) If the drug is taken every day instead of every 12 hours (1 day = 1440 minutes), the concentrations will oscillate more. The figure 7.7 shows the solution for time values for 5 days (7200 minutes):

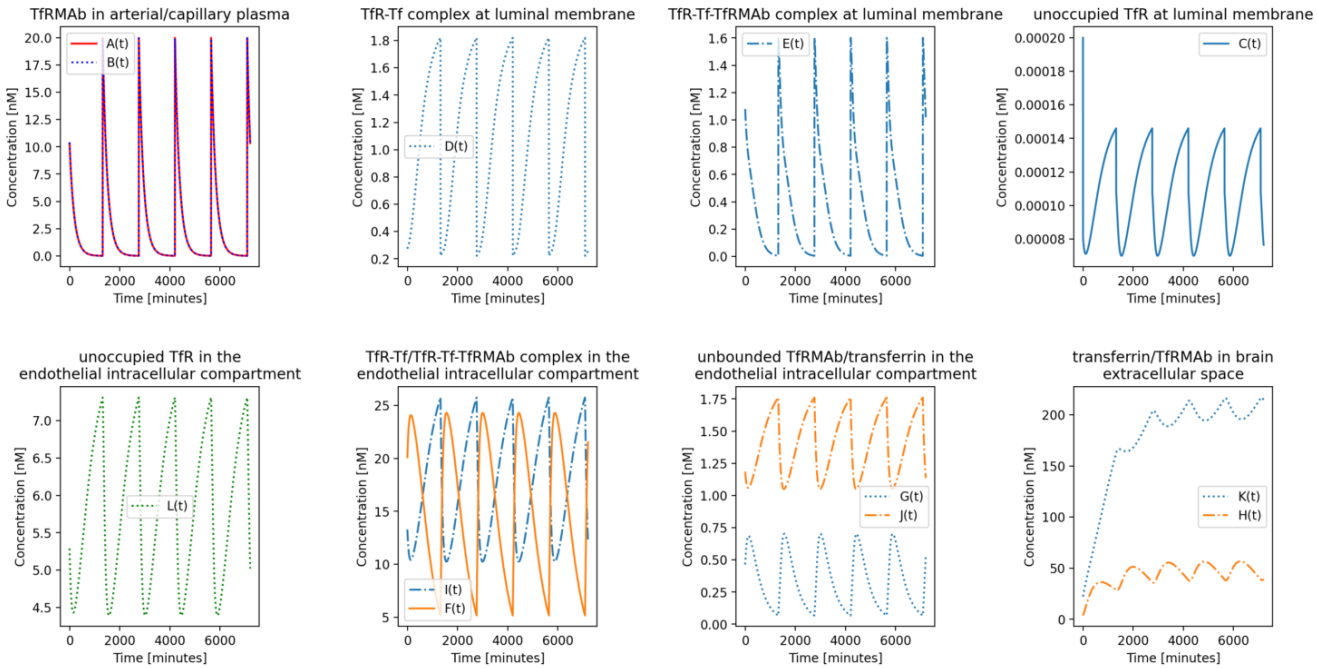


Figure 7.7: Dose 20nM is given every 24 hours (1440 minutes) for 5 days (7200 minutes)

We already have studied the long term behavior for the cases: exponential declining TfRMAb in plasma (Lemma 3) and for constant TfRMAb in plasma (Lemma 4). The periodic case of injecting the drug frequently is something between these cases: If the period is infinitely small, we get the case of constant drug injection, and if the period is infinitely long, we get the same behavior as if we had only one drug admission.
The next figure (figure 7.8) shows the same solutions as figure 7.7, but this time we add the calculated steady states from Lemma 4 (labeled as "A = 20" in figure 7.8) and the steady states from Lemma 3 (labeled "lim A to 0" in figure 7.8):

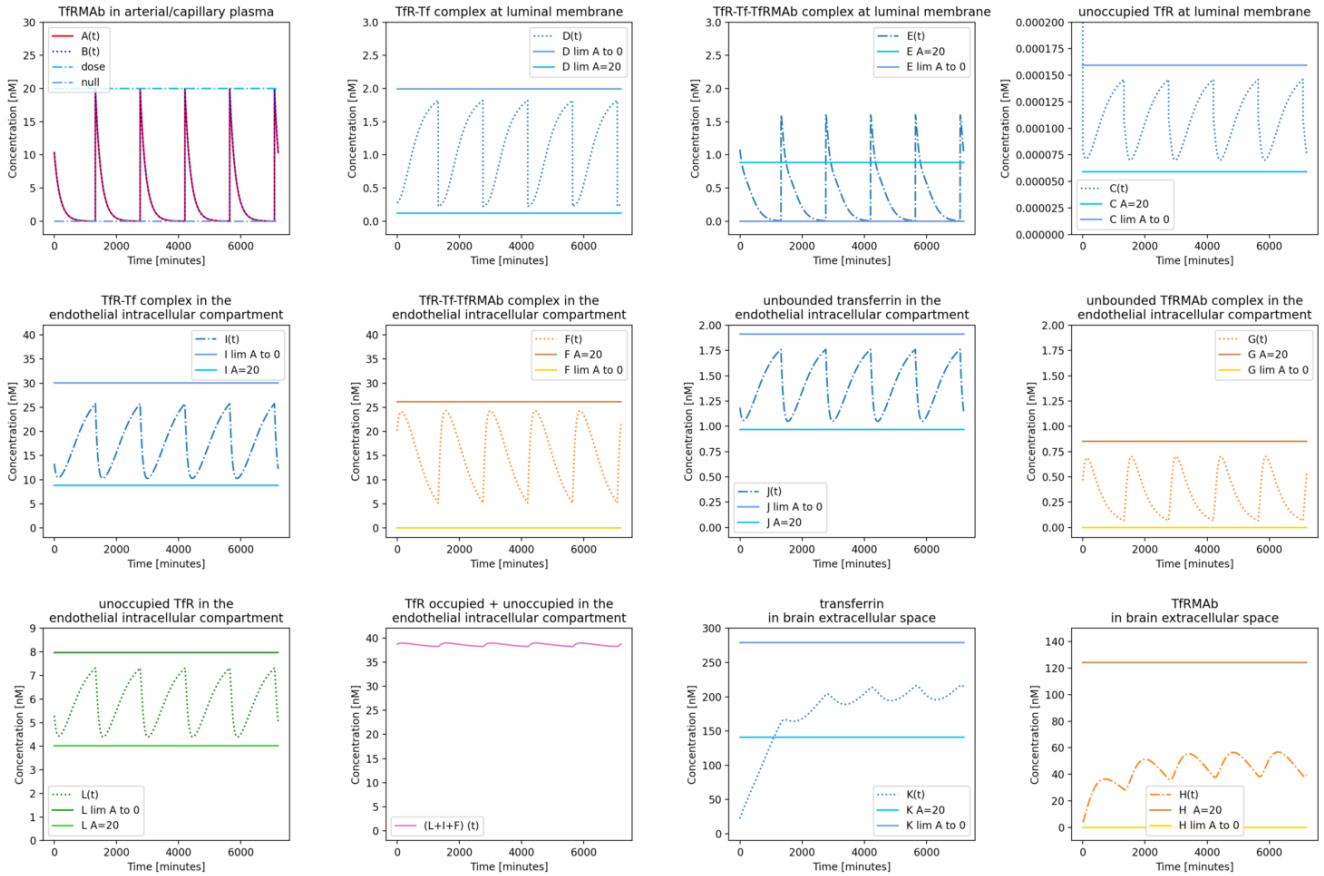


Figure 7.8: Dose 20nM is given every 24 hours (1440 minutes) for 5 days (7200 minutes) with calculated steady states

We notice, that the concentrations indeed oscillate between the calculated steady states. If not all concentration values are equal to the steady states, a single concentration term is allowed to cross those values. We see this phenomenon for the concentration of the TfR-Tf-TfRMAb complex at the luminal membrane ($E(t)$). The reason for this special behavior of the term $E(t)$ is:

Lemma 5

Let $B(t) = \bar{B}_0 = 0$ *and* $D(t) = \bar{D}_0 = 1.99$ *be the steady states from Lemma 3,* $B = \bar{B}_{20} = 20$ *and* $D = \bar{D}_{20} = 0.1198$ *be the steady states from Lemma 4, and* $\tilde{t}$ *is given so that* $D(\tilde{t}) = \bar{D}_0$ *and* $B(\tilde{t}) = \bar{B}_{20}$.

Furthermore, let $B(t) \in \left(\bar{B}_0, \bar{B}_{20}\right)$ *and* $D \in \left(\bar{D}_{20}, \bar{D}_0\right)$ $\forall t > 0$. *(this can be seen in figure 7.8)*
Then the term $k_3 D(\tilde{t}) B(\tilde{t})$ *in* $\dot{E}(t) \overset{(7.4)}{=} k_3 D(t) B(t) - k_4 E(t) - k_5 E(t)$ *is at a supremum.*

There is no need of proofing the Lemma. The result is implied directly by the assumptions. Although we do not proof, that such a $\tilde{t}$ exists, we see in the numerical solution (figure 7.8), that shortly after the injection of the drug, the system is close to such a state: B is already at 20 nM, while D is still at its peak, but will rapidly decline as soon as the TfR-Tf complex at the luminal membrane binds to the new drug molecules. This explains, why the peak values for $E(t)$ are bigger than the initial value for E.
As discussed the values for the concentrations tend to the values of Lemma 3 ($\bar{D}_0$ and $\bar{B}_0$). Injection of the drug to the plasma means, that the value of the term B suddenly rises to $\bar{B}_{20}$ (Lemma 3), while the value of the term D is still close to $\bar{D}_0$. The Lemma states, that in this situation the derivative $\dot{E}(t)$ is bigger, than for other values for B and D, that occur in the model. This explains the extreme increase of $E(t)$ after the TfRMAb injection in the model with multiple TfRMAb injections.
The next two figures show how increasing the period of drug injection diverges to the case of constant drug injection (figure 7.9), and how reducing the period of drug injection makes the concentration terms diverge to the values, which we found while studying the long term behavior of A with one initial dose (figure 7.10): (values in Lemma (figure 7.10))

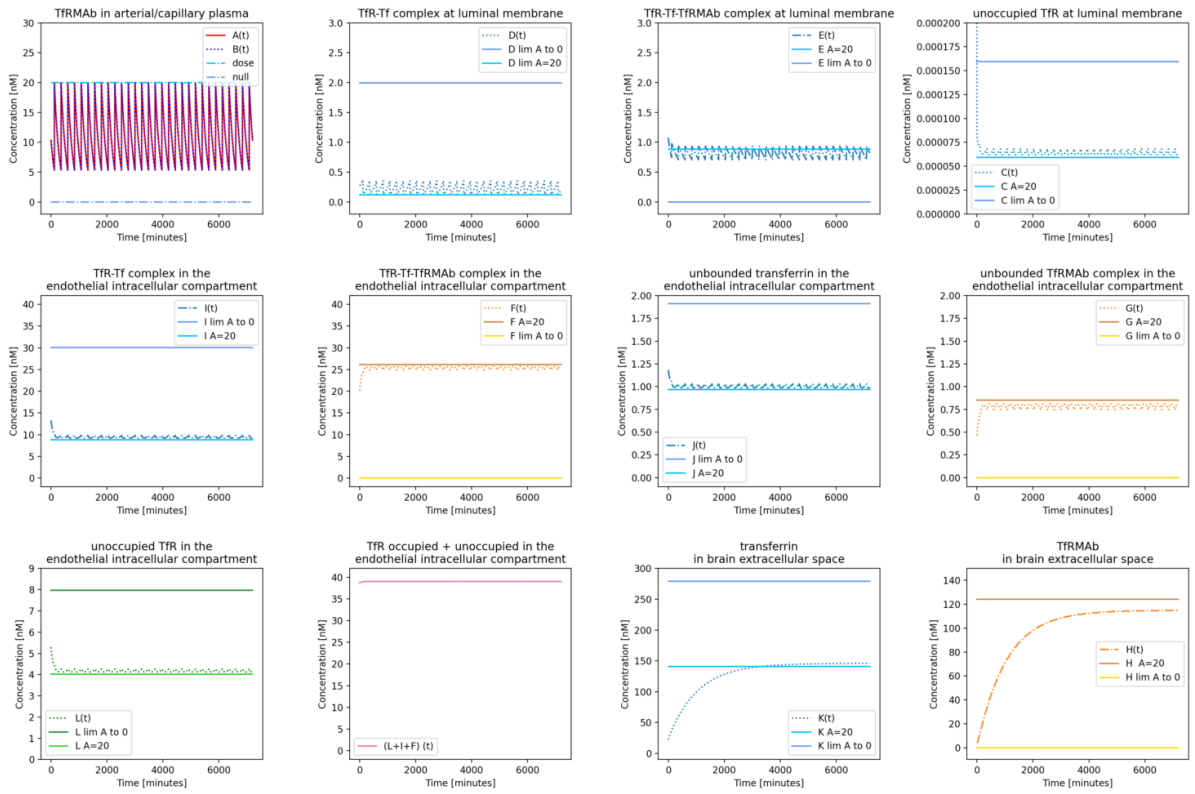


Figure 7.9: Dose 20 nM is given every 4 hours (240 minutes) for 5 days (7200 minutes)

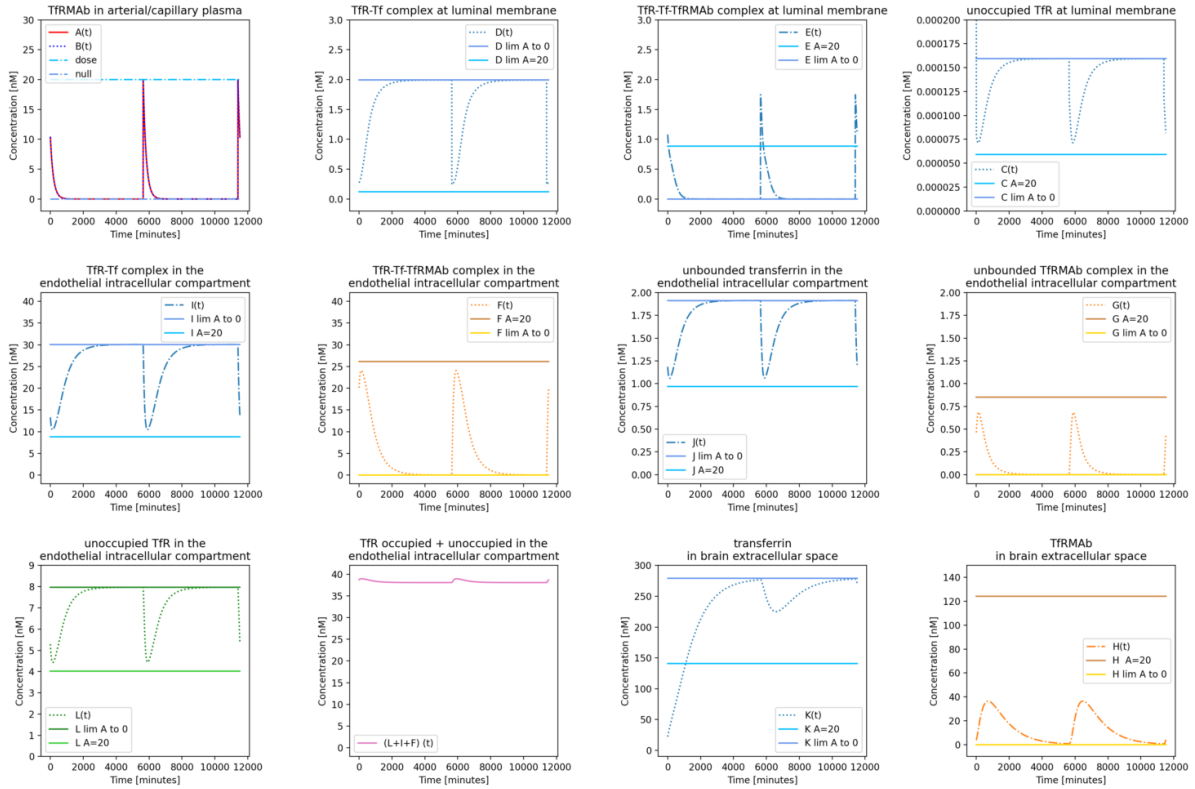


Figure 7.10: Dose 20 nM is given every 4 days (5760 minutes) for 8 days (11520 minutes)

7.3 Simplifications of the model

In this section different approaches to reduce the number of equations and parameter values are presented. We compare these approaches with the original model to evaluate their strengths and weaknesses.

7.3.1 Error function

To quantify the deviation from the solution of all equations (7.1)-(7.11) (model "Figure 7.2") to the simplified models we define the following error function:

Definition 11 (Error Function)
We define the relative error for a time interval 0 to 3000 minutes for the numerical solutions of the i-th concentration term ($i \in \{D, E, F, G, H\}$) with $sol_i(t)$ being the solution of the IVP ((7.1)-(7.11)) with the initial values from table 7.7 (model "Figure 7.2").

$$\epsilon_i(f_{model}) := \frac{2}{\int_0^{3000} f_{model}(t)\,dt + \int_0^{3000} sol_i(t)\,dt} \int_0^{3000} |f_{model}(t) - sol_i(t)|\,dt \tag{7.12}$$

Note: $f_{model}(t) > 0$ and $sol_i(t) > \forall t \in (0, 3000)$

7.3.2 Reduction of the complexity by not distinguishing between arterial and capillary plasma

The first and most obvious simplification is to reduce the system complexity by not distinguishing between TfRMAb concentrations in arterial ($A(t)$) and capillary ($B(t)$) plasma. As we have already noticed, while studying the figures 7.3 - 7.10, the graphs of A and B are almost overlapping. Biological this makes sense, since the figures display the dynamics 2 hours after the first drug injection. At this time point the drug should be already distributed in the blood. If we compare $A(t)$ and $B(t)$ by calculating the relative errors with the Error Function from Definition 11, we get the value $\epsilon = 0.000555$ for parameter values from table 7.5, table 7.6, IVs from table 7.7, $R_0 = 25000$, $A_0 = 10.34$ and $\alpha = 0.0055\frac{1}{min}$.
From now on we assume for all simplified systems, that $A(t) = B(t)$. Although $A(t)$ is given explicitly it gets its own boxes (like the other concentration terms) in the following figures for better overview.
The following simplifications will focus on the dynamics of the drug (TfRMAb) in the compartments:

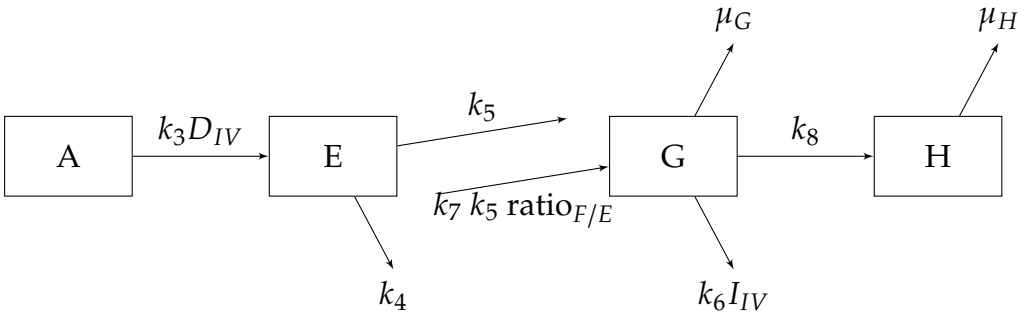


Figure 7.11: Figure showing a simplification of model 7.2 with only the following concentration variables for TfRMAb: $E(t)$, the variables for unbounded TfRMAb $G(t)$ and $H(t)$ and the variable for the initial does $A(t)$

7.3.3 Modeling the TfRMAb uptake with the four compartments (A, E, G and H)

We start with only describing the TfRMAb concentrations in the plasma ($A(t)$), in the brain extracellular space ($H(t)$) and in the BBB. In the BBB the TfRMAb appear bounded in the TfR-Tf-TfRMAb complex at the luminal membrane ($E(t)$) and unbounded in the endothelial intracellular compartment ($G(t)$). The parameter values (k_i, μ_i) are not changed. The variables, which are left out for the simplification, are replaced by the given IVs. These IVs from table 7.7 represent the concentrations 120 minutes after drug injection. We use the initial value for $D(t)$ (TfR-Tf complex at luminal membrane) to compute the TfRMAb uptake from the plasma to the luminal membrane ($k_3 D_{IV}$) and the initial value for $I(t)$ (TfR-Tf complex in the endothelial intracellular compartment) to compute the binding of the unbounded TfRMAb ($G(t)$) to the TfR-Tf complex in the endothelial intracellular compartment. We see, that we underestimate $D(t)$ by comparing the initial D_{IV} to the curve in figure 7.3. Furthermore, removing the term $F(t)$ (TfR-Tf-TfRMAb complex in the endothelial intracellular compartment) does not give the model the possibility to store the bounded TfRMAb in the endothelial intracellular compartment. The term $k_6 I_{IV} G$ gets subtracted in (7.14), but is not added to one of the other equations, meaning that the model loses those TfRMAb. These two weaknesses lead to a lower drug uptake to the brain. This can be seen by comparing the figures 7.3 and 7.12. To compensate, that the concentration of the TfR-Tf-TfRMAb complex is lower at the luminal membrane ($E(t)$) than in the endothelial intracellular compartment $F(t)$, we use the constant $\text{ratio}_{F/E} = 463.546$, which we have calculated from the numerical solutions of the model with all equations. ((7.1)-(7.11))

$$\dot{E} = k_3 D_{IV} A - k_4 E - k_5 E \tag{7.13}$$

$$\dot{G} = k_7\ k_5\ \text{ratio}_{F/E}\ E - k_6 I_{IV} G - k_8 G - \mu_G G \tag{7.14}$$

$$\dot{H} = k_8 G - \mu_H H \tag{7.15}$$

We see despite the described weaknesses, that the behavior of the curves is similar to the numerical solutions of the complete system "Figure 7.2" plotted in figure 7.3.

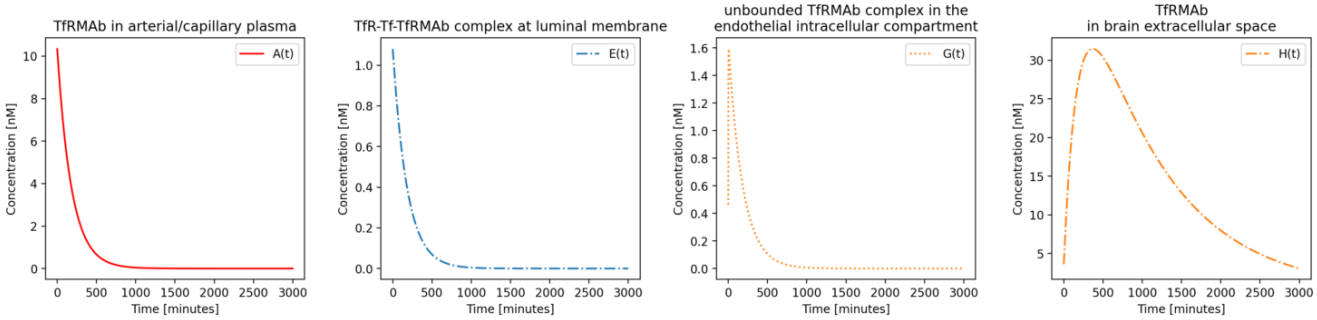


Figure 7.12: Numerical solution of the model from figure 7.11 with one initial dose (parameter values from table 7.5 for k_i, from table 7.6 for μ_i, from table 7.7 for the IVs, $A_0 = 10.34$ and $\alpha = 0.0055\frac{1}{min}$)

7.3.4 Modeling the TfRMAb uptake with the four compartments (A, E, F and H)

Like the previous model this model describes the TfRMAb uptake through the luminal membrane and the endothelial intracellular compartment. The biggest difference is, that the term $G(t)$ gets replaced by the term $F(t)$. This means, that we analyze the bounded TfRMAb instead of the unbounded TfRMAb. Broadly speaking we now have the following pipeline:
unbounded TfRMAb in the blood plasma $\longrightarrow$ bounded TfRMAb in the BBB $\longrightarrow$ unbounded TfRMAb in the brain
By ignoring the variable $I(t)$ in the term k_6IG in the equations (7.5) and (7.6) we keep the model linear. To compensate this simplification we need to find new parameter values for k_5 (in (7.4) and (7.5)), k_7 (in (7.5) and in (7.6)) and k_8 (in (7.6) and (7.7)). These three parameter values will be replaced by $\tilde{k}_5$, $\tilde{k}_7$ and $\tilde{k}_{78}$. These values can be computed by using the following trick: We know from the Lemmas 3 and 4, and from the figures 7.4 and 7.5, that the solutions tends to these calculated steady states. This leads to the following Lemma:

Lemma 6
Let:

$$\tilde{k}_5 = \dot{F}_{IV}\left(E_{IV} - \frac{F_{IV}\bar{E}_{20}}{\bar{F}_{20}}\right)^{-1}$$
$$\tilde{k}_7 = \tilde{k}_5 \cdot \frac{\bar{E}_{20}}{\bar{F}_{20}}$$
$$\tilde{k}_{78} = \frac{\mu_H \bar{H}_{20}}{\bar{F}_{20}}$$

with $\bar{E}_{20}$, $\bar{F}_{20}$ and $\bar{H}_{20}$ being the values from Lemma 4 (steady states), E_{IV} and F_{IV} being the initial values from table 7.7, and let $\dot{F}_{IV}$ be the value of term (7.5) for the initial values:
$\dot{F}_{IV} = k_5 E_{IV} + k_6 I_{IV} G_{IV} - k_7 F$
Then:

$$\begin{aligned} 0 &= \tilde{k}_5 \bar{E}_{20} - \tilde{k}_7 \bar{F}_{20} \\ \dot{F}_{IV} &= \tilde{k}_5 E_{IV} - \tilde{k}_7 F_{IV} \\ 0 &= \tilde{k}_{78} \bar{F}_{20} - \mu_H \bar{H}_{20} \end{aligned}$$

Proof:

$$\begin{aligned} \tilde{k}_5 \bar{E}_{20} - \tilde{k}_7 \bar{F}_{20} &= \tilde{k}_5 \bar{E}_{20} - \tilde{k}_5 \cdot \frac{\bar{E}_{20}}{\bar{F}_{20}} \bar{F}_{20} \\ &= \tilde{k}_5 \bar{E}_{20} - \tilde{k}_5 \bar{E}_{20} \\ &= 0 \end{aligned}$$

$$\begin{aligned} \tilde{k}_5 E_{IV} - \tilde{k}_7 F_{IV} &= \frac{\dot{F}_{IV}}{\left(E_{IV} - \frac{F_{IV}\bar{E}_{20}}{\bar{F}_{20}}\right)} E_{IV} - \frac{\dot{F}_{IV}}{\left(E_{IV} - \frac{F_{IV}\bar{E}_{20}}{\bar{F}_{20}}\right)} \cdot \frac{\bar{E}_{20}}{\bar{F}_{20}} F_{IV} \\ &= \dot{F}_{IV} \left(\frac{\bar{F}_{20} E_{IV}}{E_{IV}\bar{F}_{20} - F_{IV}\bar{E}_{20}} - \frac{\bar{F}_{20}}{E_{IV}\bar{F}_{20} - F_{IV}\bar{E}_{20}} \cdot \frac{\bar{E}_{20}}{\bar{F}_{20}} F_{IV} \right) \\ &= \dot{F}_{IV} \left(\frac{\bar{F}_{20} E_{IV}}{E_{IV}\bar{F}_{20} - F_{IV}\bar{E}_{20}} - \frac{\bar{E}_{20} F_{IV}}{E_{IV}\bar{F}_{20} - F_{IV}\bar{E}_{20}} \right) \\ &= \dot{F}_{IV} \frac{E_{IV}\bar{F}_{20} - F_{IV}\bar{E}_{20}}{E_{IV}\bar{F}_{20} - F_{IV}\bar{E}_{20}} \\ &= \dot{F}_{IV} \end{aligned}$$

$$\begin{aligned} \tilde{k}_{78} \bar{F}_{20} - \mu_H \bar{H}_{20} &= \frac{\mu_H \bar{H}_{20}}{\bar{F}_{20}} \bar{F}_{20} - \mu_H \bar{H}_{20} \\ &= \mu_H \bar{H}_{20} - \mu_H \bar{H}_{20} \\ &= 0 \end{aligned}$$

■

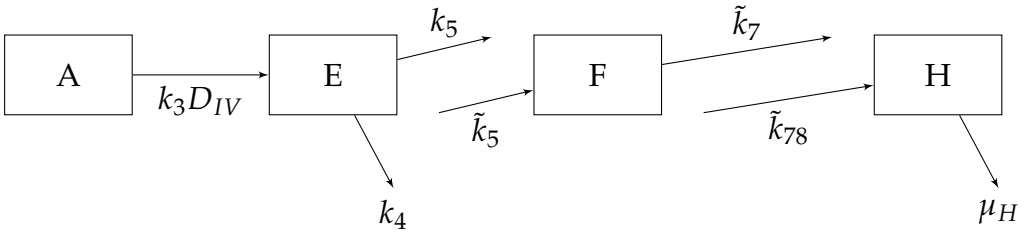


Figure 7.13: Figure showing a simplification of model 7.2 with only the following concentration variables for TfRMAb: initial dose $A(t)$, TfRMAb in the brain $H(t)$ and the variables for bounded TfRMAb in the BBB: $E(t)$ and $F(t)$

The dynamics are shown in figure 7.13 and the ODEs ((7.16)-(7.18)).

$$\dot{E} = k_3 D_{IV} A - k_4 E - k_5 E \tag{7.16}$$
$$\dot{F} = \tilde{k}_5 E - \tilde{k}_7 F \tag{7.17}$$
$$\dot{H} = \tilde{k}_{78} F - \mu_H H \tag{7.18}$$

We see, that the $\dot{E}$ term ((7.13) versus (7.16)) did not change. The other terms ((7.17) and (7.18)) use the values calculated in Lemma 6.
The following figure 7.14 shows, that this rather easy approach already gives us a good overview of the dynamics:

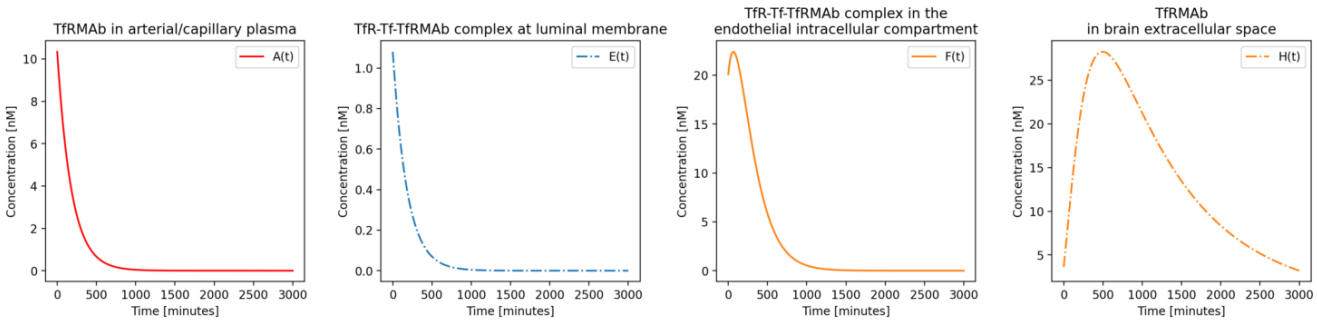


Figure 7.14: Numerical solution of the model from figure 7.13 with one initial dose (parameter values from table 7.5 for k_i, from Lemma 6 for $\tilde{k}_i$, from table 7.6 for μ_i, from table 7.7 for the IVs, $A_0 = 10.34$ and $\alpha = 0.0055 \frac{1}{min}$)

7.3.5 Modeling the TfRMAb uptake via the concentration of the TfR-Tf complex at the endothelial luminal membrane ($D(t)$)

We use the logic of the previous model, but add the term for the concentration of the TfR-Tf complex at the endothelial luminal membrane ($D(t)$) to the model. Since the endothelial luminal membrane is an essential part of the BBB, and since we are actually interested in dynamics

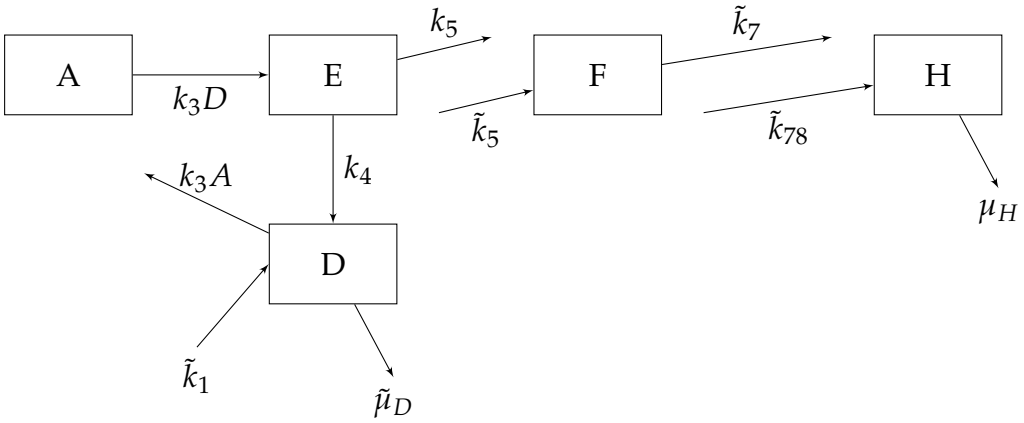


Figure 7.15: Figure showing model 7.13 with one extra variable: $D(t)$

of the drug binding to transferrin during the uptake, this is an essential improvement.

$$\dot{D} = \tilde{k}_1 - \tilde{\mu}_D D - k_3 D A + k_4 E \tag{7.19}$$
$$\dot{E} = k_3 D A - k_4 E - k_5 E \tag{7.20}$$
$$\dot{F} = \tilde{k}_5 E - \tilde{k}_7 F \tag{7.21}$$
$$\dot{H} = \tilde{k}_{78} F - \mu_H H \tag{7.22}$$

We already calculated the values $\tilde{k}_5$, $\tilde{k}_7$ and $\tilde{k}_{78}$ in Lemma 6. For the equation (7.19) we need to find the values for $\tilde{k}_1$ and $\tilde{\mu}_D$.

Lemma 7
Let:

$$\tilde{\mu}_D = \frac{k_3 \bar{D}_{20} \bar{A}_{20} - k_4 \bar{E}_{20}}{\bar{D}_0 - \bar{D}_{20}}$$
$$\tilde{k}_1 = \tilde{\mu}_D \bar{D}_0$$

With $\bar{D}_0$ and $\bar{E}_0$ being the values (steady states) from Lemma 3, $\bar{D}_{20}$ and $\bar{E}_{20}$ being the values (steady states) from Lemma 4, and $\bar{A}_0 = 0$ and $\bar{A}_{20} = 20$ ($A = 0$ was the assumption for Lemma 3 and $A = 20$ was the assumption for Lemma 4)
Then:

$$0 = \tilde{k}_1 - \tilde{\mu}_D \bar{D}_0 - k_3 \bar{D}_0 \bar{A}_0 + k_4 \bar{E}_0$$
$$0 = \tilde{k}_1 - \tilde{\mu}_D \bar{D}_{20} - k_3 \bar{D}_{20} \bar{A}_{20} + k_4 \bar{E}_{20}$$

Proof:

$$
\begin{aligned}
&\tilde{k}_1 - \tilde{\mu}_D \bar{D}_0 - k_3 \bar{D}_0 \bar{A}_0 + k_4 \bar{E}_0 \\
&= \tilde{\mu}_D \bar{D}_0 - \tilde{\mu}_D \bar{D}_0 - k_3 \bar{D}_0 0 + k_4 0 \\
&= 0 \\
&\tilde{k}_1 - \tilde{\mu}_D \bar{D}_{20} - k_3 \bar{D}_{20} \bar{A}_{20} + k_4 \bar{E}_{20} \\
&= \tilde{\mu}_D \bar{D}_0 - \tilde{\mu}_D \bar{D}_{20} - k_3 \bar{D}_{20} \bar{A}_{20} + k_4 \bar{E}_{20} \\
&= \tilde{\mu}_D \left(\bar{D}_0 - \bar{D}_{20} \right) - k_3 \bar{D}_{20} \bar{A}_{20} + k_4 \bar{E}_{20} \\
&= \frac{k_3 \bar{D}_{20} \bar{A}_{20} - k_4 \bar{E}_{20}}{\bar{D}_0 - \bar{D}_{20}} \left(\bar{D}_0 - \bar{D}_{20} \right) - k_3 \bar{D}_{20} \bar{A}_{20} + k_4 \bar{E}_{20} \\
&= k_3 \bar{D}_{20} \bar{A}_{20} - k_4 \bar{E}_{20} - k_3 \bar{D}_{20} \bar{A}_{20} + k_4 \bar{E}_{20} \\
&= 0
\end{aligned}
$$

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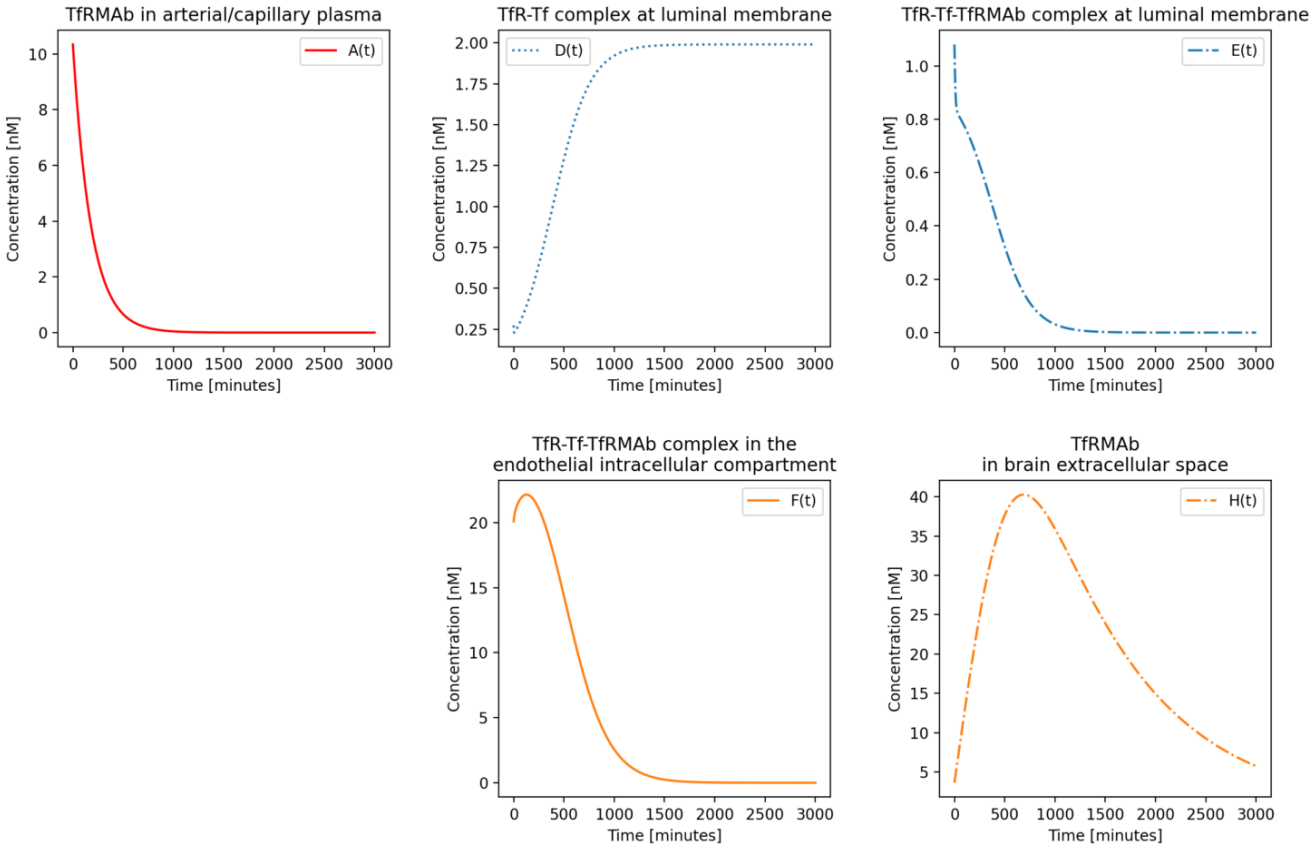


Figure 7.16: Numerical solution of the model "Figure 7.15" with one initial dose (parameter values from table 7.5 for k_i, from Lemma 6 and from Lemma 7 for $\tilde{k}_i$, from table 7.6 for μ_i, from Lemma 7 for $\tilde{\mu}_D$, from table 7.7 for the IVs, $A_0 = 10.34$ and $\alpha = 0.0055 \frac{1}{min}$)

7.3.6 Comparison of the models "Figure 7.11", "Figure 7.13" and "Figure 7.15"

To evaluate the quality of the simplifications of the reference model "Figure 7.2", we compare the numerical solutions for the parameter values from table 7.5 for k_i, from Lemma 6 (for the models shown in figure 7.13 and in figure 7.15) and from Lemma 7 for $\tilde{k}_i$ (for the model shown in figure 7.15), from table 7.6 for μ_i, from Lemma 7 for $\tilde{\mu}_D$ (for the model shown in figure 7.15), from table 7.7 for the IVs, for $A_0 = 10.34$ and for $\alpha = 0.0055\frac{1}{min}$ (for exponential declining $A(t)$ (Definition 6.1.)) We assume exponential decline $A(t)$ unless otherwise stated.
Note: The initial model displayed in figure 7.2, with the numerical solutions displayed in figure 7.3 and with the ODEs (7.1)-(7.11) is the benchmark for the simplified versions, in the error function (Definition 11) the numerical solutions of the model are also referred to as $\text{sol}_i(t)$. A tabular overview of the models in this chapter can be found in the Appendix in section 9.3.

Relative errors

We use the Error Function, which we defined in Definition 11, to quantify the relative errors of the simplified models shown in the figures 7.11, 7.13 and 7.15 from the sections 7.3.3, 7.3.4 and 7.3.5:

Model (f_{model})	Figure 7.11	Figure 7.13	Figure 7.15
Section	7.3.3	7.3.4	7.3.5
Linear?	Yes	Yes	No
$\epsilon_i(f_{\text{model}})$ for $i = D$	not defined	not defined	0.06754
$\epsilon_i(f_{\text{model}})$ for $i = E$	0.5636	0.5636	0.1205
$\epsilon_i(f_{\text{model}})$ for $i = F$	not defined	0.8922	0.4724
$\epsilon_i(f_{\text{model}})$ for $i = G$	0.8391	not defined	not defined
$\epsilon_i(f_{\text{model}})$ for $i = H$	0.4796	0.4447	0.09742

Table 7.1: Errors of the simplifications of the normal model compared to the original model

We notice, that the model "Figure 7.15" shows a significant lower deviation from the benchmark, than the other models.

Properties of TfRMAb concentatrion in the brain $H(t)$

Since we are interested in the TfRMAb uptake to the brain, the next table shows the peak TfRMAb concentration in brain extracellular space ($H(t)$) and the time point the peak occurs:

Model	unit	$\text{sol}_H(t)$ or Figure 7.2	Figure 7.11	Figure 7.13	Figure 7.15
H peak concentration	nM	36.37	31.48	28.25	40.28
H peak time	min	718	370	495	689

Table 7.2: Maximal TfRMAb concentration in brain extracellular space ($H(t)$) in comparison

We see, that the more complex the model gets, the more time it takes for $H(t)$ to reach its peak. This absolutely makes sense, since the more variables we have, and the more compartments we have, the longer it takes for the TfRMAb to pass the BBB. Furthermore, we see, that the values from model "Figure 7.15" are closest to the values from the reference model "Figure 7.2" ($\text{sol}_H(t)$).

Long term behavior

Since the steady states, which we found in Lemma 3 and Lemma 4, were essential for setting up the simplified models, we will now check, if these simplified models actually converge to these. It is important to note, that we never have proven, that the solutions converge to them or if they are asymptotically stable.
Fist we calculate the limits for the case, that $A(t)$ tends to 0 (like in Lemma 3):

Model	$\text{sol}_i(t)$ or Figure 7.2	Figure 7.11	Figure 7.13	Figure 7.15
D lim A = 0	1.993	not defined	not defined	1.990
E lim A = 0	0	0	0	0
F lim A = 0	0	not defined	0	0
G lim A = 0	0	0	not defined	not defined
H lim A = 0	0	0	0	0

Table 7.3: Long term behavior of the different models for $A(t)$ declining exponentially

We see, that the long term behavior of all simplified modles is equivalent to the model "Figure 7.2" (reference model), if we assume that only one TfRMAb dose is taken (Lemma 3). In long term the concentration variables for TfRMAb $E(t)$, $F(t)$, $G(t)$ and $H(t)$ decline to 0. This is biological meaningful, since the one initial drug dose need to get decomposed somewhere in the system.

Model	$\text{sol}_i(t)$ or Figure 7.2	Figure 7.11	Figure 7.13	Figure 7.15
D lim A = 20	0.1198	not defined	not defined	0.1197
E lim A = 20	0.8870	2.0296	2.0296	0.8870
F lim A = 20	26.1728	not defined	59.8885	26.1731
G lim A = 20	0.8517	3.070299	not defined	not defined
H lim A = 20	124.21	447.75	284.21	124.21

Table 7.4: Long term behavior of the different models for $A(t)$ being constant at 20nM

We see, that for $A(t)$ being constant at $A_0 = 20$ nM (Definition 6.2.) the limits of the long term behavior for the model "Figure 7.15" match with the those for the reference model "Figure 7.2". This makes sense, since these values for the steady states from Lemma 3 and from Lemma 4 were used to find the values for $\tilde{k}_5$, $\tilde{k}_7$ and $\tilde{k}_{78}$ (Lemma 6), and for $\tilde{\mu}_D$ and $\tilde{k}_1$ (Lemma 7) in the equations (7.19)-(7.22) for the model "Figure 7.15". The other models "Figure 7.11" and "Figure 7.13" are too simple, and do not fulfill this property.

7.3.7 Quasi steady states

The problem with the simplifications, which we already found is: To find the new parameter values, we used Lemma 3 and Lemma 4, but these two lemmas were proven by investigating the long term behavior and using these limits to proof, that there are actually steady states. For these two proofs we used the parameter values from table 7.5 and from table 7.6. But what happens, if these parameters change? To handle this issue we use the steady states from Theorem 7, which is more general than the lemmas. This results in models, which are call quasi steady state models. This means, we keep some ODEs, but replace some others by functions, which we get from the steady states. The following example illustrates this approach:
Let this be an ODE system:

$$\dot{X} = f(X, Y)$$
$$\dot{Y} = g(X, Y)$$

We assume X_Q to be in a steady state:

$$\dot{X}_Q = 0 \implies f(X_Q, Y) = 0 \overset{\text{Solve for } X_Q}{\rightsquigarrow} X_Q = h(Y)$$

And now we have the reduced the system to one ODE:

$$\dot{Y} = g(X_Q, Y) = g(h(Y), Y)$$

This can now be solved for the initial value $y_0 = Y(0)$. (If a solution exists.)

In the following models in the section 7.3.7 we use these notations: The variables, which are assumed to be in a quasi steady state, get labeled with the letter "Q" (as in the example) and are represented by circles in figure 7.17. The variable defined by the ODEs and the initial variable $A(t)$ (Definition 6) get boxes in figure 7.17.

Quasi steady state model for $D_Q(t)$, $I_Q(t)$ and $L_Q(t)$

Let this be the ODE system:

$$\dot{C} = k_{13}L_Q(C) - k_1 C R_0 + k_2 D_Q(C) \tag{7.23}$$
$$\dot{E} = k_3 D_Q(C) A - k_4 E - k_5 E \tag{7.24}$$
$$\dot{F} = k_5 E + k_6 I_Q(C) G - k_7 F \tag{7.25}$$
$$\dot{G} = k_7 F - k_6 I_Q(C) G - k_8 G - \mu_G G \tag{7.26}$$
$$\dot{H} = k_8 G - \mu_H H \tag{7.27}$$

Using Theorem 7 we get the following variables:

$$D_Q(C) = \lambda_1 C \tag{7.28}$$
$$I_Q(C) = \frac{1}{k_{11}}\left(k_9\lambda_1 C + k_{10}\lambda_2\lambda_3 C^2\right) \tag{7.29}$$
$$L_Q(C) = \lambda_2 C \tag{7.30}$$
$$\lambda_1 = \frac{k_1 R_0}{k_2 + k_9} \tag{7.31}$$
$$\lambda_2 = \frac{1}{k_{13}}\left(k_1 R_0 - k_2\lambda_1\right) \tag{7.32}$$
$$\lambda_3 = \frac{k_9}{k_{12} + \mu_J}\lambda_1 \tag{7.33}$$

The following figure 7.17 displays the quasi steady state model:

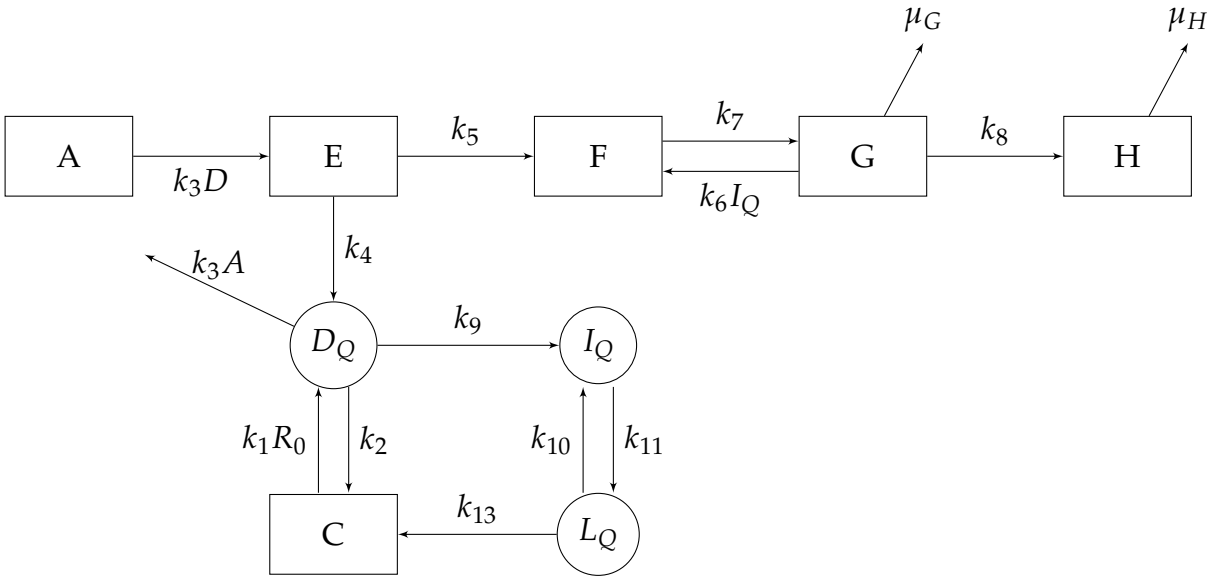


Figure 7.17: Quasi steady state model for $D_Q(t)$, $I_Q(t)$ and $L_Q(t)$ with the ODE variables for all TfRMAb concentrations (first row) and $C(t)$

The following figure shows the numerical solution of this model for one initial dose, parameter values from table 7.5, table 7.6, IVs from table 7.7, R_0 = 25000, A_0 = 10.34 and α = 0.0055$\frac{1}{min}$. The initial value for $C(t)$ was changed to 0.00002192 nM for the following reason: The initial value for $C(t)$ needs to be as accurate as possible. Since the variables interacting with $C(t)$ are in the quasi steady state, the variable $C(t)$ is not as volatile as in the reference model "Figure 7.2".
λ_1 = 12500 for the parameter values from table 7.5, table 7.6 and the initial value for $D(t)$ is 0.274 nM. So we get $\frac{0.274nM}{12500}$ = 0.00002192nM as IV for $C(t)$.

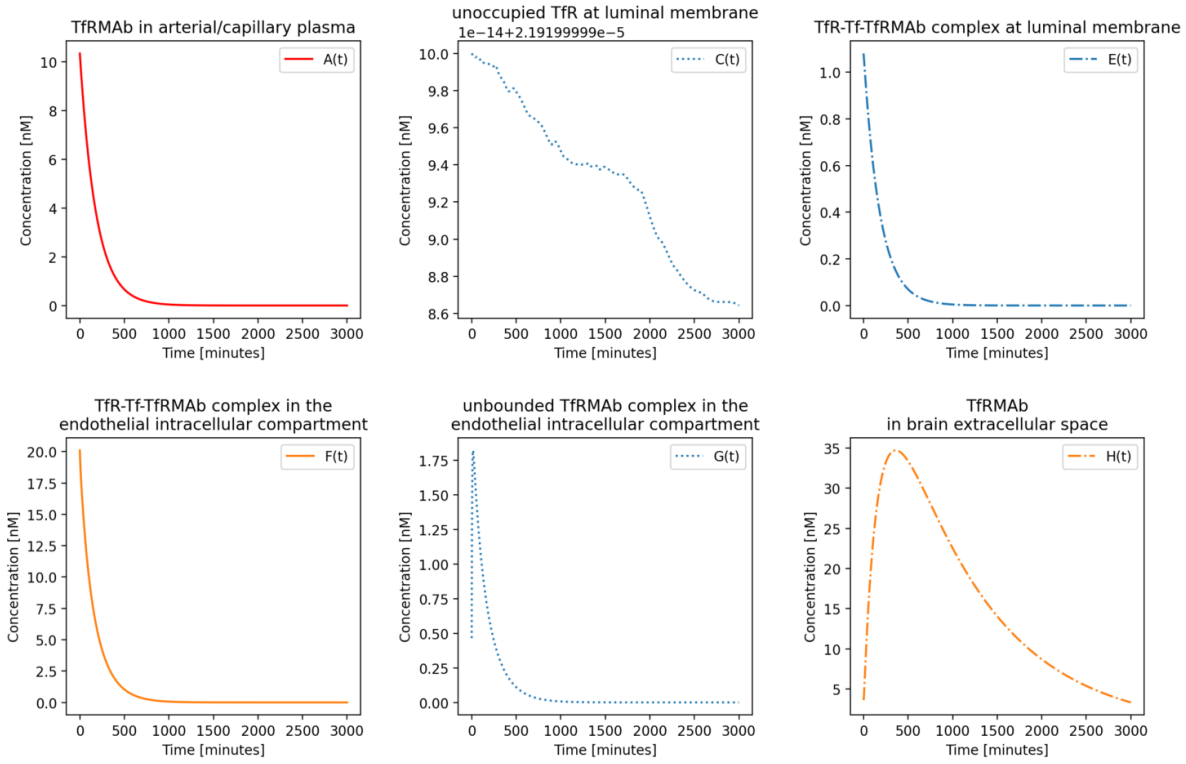


Figure 7.18: Solution of the model 7.17 with the parameter values as described above

Note, that the scale for the variable $C(t)$ "unoccupied transferrin receptor at luminal membrane" is different to the other scales. (1 unit on the y-axis = 10^{-14} nM) This implies, that the variable is almost constant.
In general the solution shown in figure 7.18 shows similar properties to the solution of the reference model ("Figure 7.3"), but like in the other models (e.g. "Figure 7.11)" the peak of the $H(t)$ occurs at a too early time point.

7.3.8 Parameter dependency of the concentration variables for a fixed time point

The next figure 7.19 shows the values for the concentration variables $C(t)$, $D(t)$, $E(t)$, $F(t)$, $G(t)$ and $H(t)$ for t=720 min (12 hours) for different parameter values for k_3 (TfRMAb association with Tf-TfR complex at plasma membrane) and k_6 (TfRMAb association with Tf-TfR complex within endothelial cell). We assume $k_3 = k_6$. It makes sence to simulate, how the concentration variables depend on these parameter values, since the binding of TfRMAb to the Tf-TfR complex is the essential step for the TfRMAb to pass the BBB. (Compare this with chapter 6, where we modeled the dependencies of the concentration terms for the values of the transferrin affinity.) The other parameters are the same as those chosen to calculate the numerical solution of model "Figure 7.17" shown in figure 7.18. The only exception is again the initial value for $C(t)$. We compare these three models:

The reference model "Figure 7.2" is labeled as "sol all", the model "Figure 7.15" is labeled as "sol for E, F, H and D", and the quasi steady state model "Figure 7.17" is labeled as "QSS". The two differences between the two curves labeled with "QSS" are primarily the chosen IVs for $C(t)$ and secondarily for the curve with the label "QSS for C const" the equation for $\dot{C}$ (7.23) was set to 0. As we see in the top right subplot of both figures, this makes no difference: "QSS for C const" and "QSS for C var" (the solution of the model with the equation (7.23) not being set to 0) seem to be parallel. What actually does matter for the behavior in the simulations is the selection of the IV.

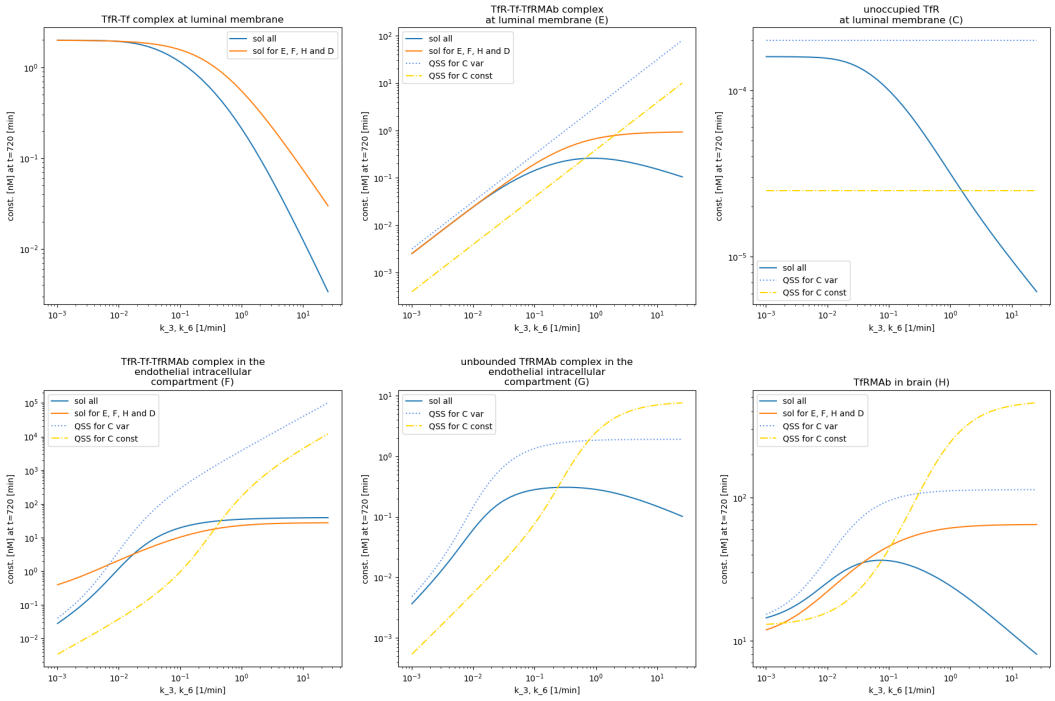


Figure 7.19: These curves show the concentrations for a constant t=720 min, but for variable k_3=k_6

7.3.9 Evaluation of the simplifications

In the appendix in section 9.3 is a tabular overview of all the models in this chapter, were the ODEs and the figures are referred to.
To summarize the results from the different approaches to simplify the reference model "Figure 7.2" we can state the obvious, that none of these new models seems to be perfect. All found models have strengths and weaknesses. We have already compared the models "Figure 7.11", "Figure 7.13" and "Figure 7.15" in section 7.3.6 with the reference model, where we investigated the relative errors and the long term behavior. The result was, that model "Figure 7.15" shows very low relative errors and the same long term behavior as the reference model. Although this model is simpler than the quasi steady state model, it seems to be capable of simulating variable parameter values for the TfRMAb association with the Tf-TfR complex (k_3 and k_6).

These results (figure 7.19) are even better than those for the steady state model. (figure 7.19) Note that k_3 is a model parameter in the ODEs (7.19) and (7.20), but k_6 is not a parameter in the model. This means, that this statement is only true under the assumption $k_3 = k_6$.
To summarize these findings: Adding the concentration variable $D(t)$ to the ODEs, as we did for the model "Figure 7.15" (7.19), seems to improve the model significantly, but adding more variables as quasi steady states does not seem to improve the model.

7.4 Data

	Description	Value	Unit
k_0	brain capillary blood flow	42	1/min
k_1	Tf association with TfR at endothelial plasma membrane	0.1	1/(nM*min)
k_2	Tf dissociation from TfR at endothelial plasma membrane	0.06	1/min
k_3	TfRMAb association with Tf-TfR complex at plasma membrane	0.06	1/(nM*min)
k_4	TfRMAb dissociation with Tf-TfR complex at plasma membrane	0.022	1/min
k_5	internalization of TfRMAb-Tf-TfR complex from E to F	0.14	1/min
k_6	TfRMAb association with Tf-TfR complex within endothelial cell	0.06	1/(nM*min)
k_7	TfRMAb dissociation with Tf-TfR complex within endothelial cell	0.022	1/min
k_8	MAb exocytosis into brain interstitium	0.14	1/min
k_9	internalization of Tf-TfR complex from endothelial luminal membrane into intra-endothelial compartment	0.14	1/min
k_{10}	Tf association with TfR within endothelial cell	0.1	1/(nM*min)
k_{11}	Tf dissociation from TfR within endothelial cell	0.06	1/min
k_{12}	Tf exocytosis into brain interstitium	0.14	1/min
k_{13}	intracellular TfR recycling back to plasma membrane	0.035	1/min

Table 7.5: Parameter values for k_i

	Description	Value	Unit
μ_B	TfRMAb removal from B other than binding to endothelial luminal membrane	0.00096	1/min
μ_H	degradation of TfRMAb in brain	0.00096	1/min
μ_K	degradation of Tf in brain	0.00096	1/min
μ_J	degradation of free Tf within endothelial compartment	0.0058	1/min
μ_G	degradation of free TfRMAb within endothelial compartment	0.0058	1/min

Table 7.6: Parameter values for μ_i

Concentration	Description	IV	Unit
B	TfRMAb in capillary plasma	10.3	nM
C	unoccupied TfR at brain endothelial luminal membrane	0.0002	nM
D	TfR-Tf complex at endothelial luminal membrane	0.274	nM
E	TfR-Tf-TfRMAb complex at endothelial luminal membrane	1.08	nM
F	TfR-Tf-TfRMAb complex in endothelial intracellular compartment	20.1	nM
G	unbounded TfRMAb in endothelial intracellular compartment	0.464	nM
H	TfRMAb in brain extracellular space	3.62	nM
I	TfR-Tf complex in endothelial intracellular compartment	13.3	nM
J	unbounded Tf in endothelial intracellular compartment	1.18	nM
K	Tf in brain extracellular space	22.2	nM
L	unoccupied TfR in endothelial intracellular compartment	5.29	nM

Table 7.7: IVs for model 7.2 representing the concentration of the variables 2 hours after initial dose of 0.2 mg/kg

8 Discussion and Outlook

The different approaches to model the uptake through the BBB of the bispecific antibodies in chapter 6 and the uptake of the monoclonal antibodies in chapter 7, which both target the transferrin receptor, give us a good intuition and understanding of the dynamics and variable concentrations within the blood plasma, the BBB and the brain. By simplifying the model in chapter 7 we reduce not only the number of variables, but also the number of model parameters. Doing this can improve the meaningfulness of the model, if less parameter values are known. On the other hand, the reduced model is not capable of showing all dynamics in detail and simulating variations of the parameters left out.
In both models the effect of changing the parameter values on the variables can be investigated further. We have studied the dependencies of the concentration variables on the most important parameter values, but there are multiple more options. A sensitivity analysis could also be done. The figures showing the dependencies (e.g. figure 7.19) plot the concentrations as functions with the parameter values, not with the time, as variables, but it might be interesting to make time and parameter values both variable. This would give us the possibility to express these properties as functions of variable parameter values: the time point when the function reaches a maximum, the time point when the concentration hits a certain threshold and more. A next possible step could be to find optimal parameter values for those conditions.
Since the model in chapter 7 does not only describe the uptake of the antibodies like the model in chapter 7, but also the effect of the drug on the amyloid-β peptides in the brain and the effect of the drug on reticulocyte in the plasma, it might be an interesting aspect to add a variable for the effect of these antibodies in the brain as well.
In the book by Caspar and Hungerbühler, 2022 [3] the approach of optimal control with Laplace Transformation in pharmacokinetics models is discussed. These models are simpler than the models we have discussed, but the possibility of optimizing wanted properties with this approach might be a topic of further research for the more complex models of antibody uptake to the brain.
In the article by W. Pardridge and T. Chou, 2021 [13] the dynamics of antibodies targeting the insulin receptor are also described. It might be interesting to try a similar approach to simplify this model as well, as we have done with the TfR model.

9 Appendix

9.1 List of abbreviations

ODE	ordinary differential equation
IV	initial value
IVP	initial value problem
PK	pharmacokinetics
BBB	blood brain barrier
Tf	transferrin
TfR	transferrin receptor
LC-MS/MS	liquid chromatography/mass spectroscopy
BPR	blood plasma ratio
BDF	backward differentiation formula
APP	amyloid precursor protein
MAb	monoclonal antibodies
TfRMAb	transferrin receptor monoclonal antibodies
RMT	receptor medicated transport

9.2 List of global variables

$A(t)$	concentration of the drug in the blood plasma
$P(t)$	concentration of the drug in the periphery compartment
$H(t)$	concentration of the drug in the brain

9.3 List of models in chapter 7

Model	ODE System	Numerical Solution	Variables	Note
Figure 7.2	(7.1)-(7.11)	Figure 7.3	$B(t)$-$L(t)$ and initial dose $A(t)$	vanilla model, which is the benchmark for the simplified versions, in the error function (Definition 11) it is also referred to as $\text{sol}_i(t)$
Figure 7.11	(7.13)-(7.15)	Figure 7.12	$E(t)$, $G(t)$ and $H(t)$ and initial dose $A(t)$	
Figure 7.13	(7.16)-(7.18)	Figure 7.14	$E(t)$, $F(t)$ and $H(t)$ and initial dose $A(t)$	
Figure 7.15	(7.19)-(7.22)	Figure 7.16	$D(t)$, $E(t)$, $F(t)$ and $H(t)$ and initial dose $A(t)$	
Figure 7.17	(7.23)-(7.27)	Figure 7.18	$C(t)$, $E(t)$, $F(t)$, $G(t)$ and $H(t)$ and initial dose $A(t)$	$D_Q(t)$, $I_Q(t)$ and $L_Q(t)$ are the quasi steady state variables

9.4 Proof of Lemma 4 in chapter and section 7.2.2 with the not rounded values

- $\bar{B}$ = 19.99658626735864
- $\bar{C}$ = $5.9253258349709386e \cdot 10^{-5}$
- $\bar{D}$ = 0.11976548876967165
- $\bar{F}$ = 26.17281262518254
- $\bar{G}$ = 0.8517150076840816
- $\bar{H}$ = 124.20843862052891

- $\bar{I}$ = 8.837498959143081
- $\bar{J}$ = 0.9667161628812981
- $\bar{K}$ = 140.97944042012693
- $\bar{L}$ = 4.027063329945519

Proof:

$$\dot{B} \overset{(7.1)}{=} k_0\left(A-\bar{B}\right)-k_3\bar{D}\bar{B}-\mu_B\bar{B}+k_4\bar{E}$$
$$= 42*(20-19.99658626735864)-0.06*0.11976548876967165*19.99658626735864$$
$$-0.022*0.887000343716708+0.022*0.887000343716708=0$$

$$\dot{C} \overset{(7.2)}{=} k_{13}\bar{L}-k_1\bar{C}R_0+k_2\bar{D}$$
$$= 0.035*4.027063329945519-0.1*0.000059253258349709386*25000$$
$$+0.022*0.11976548876967165=0$$

$$\dot{D} \overset{(7.3)}{=} k_1\bar{C}R_0-k_2\bar{D}-k_9\bar{D}-k_3\bar{D}\bar{B}+k_4\bar{E}$$
$$= 0.1*0.000059253258349709386*25000-0.022*0.11976548876967165$$
$$-0.14*0.11976548876967165-0.06*0.11976548876967165*19.99658626735864$$
$$+0.022*0.887000343716708=0$$

$$\dot{E} \overset{(7.4)}{=} k_3\bar{D}\bar{B}-k_4\bar{E}-k_5\bar{E}$$
$$= 0.06*0.11976548876967165*19.99658626735864-0.022*0.887000343716708$$
$$-0.14*0.887000343716708=0$$

$$\dot{F} \overset{(7.5)}{=} k_5\bar{E}+k_6\bar{I}\bar{G}-k_7\bar{F}$$
$$= 0.14*0.887000343716708+0.06*8.837498959143081*0.8517150076840816$$
$$-0.022*26.17281262518254=0$$

$$\dot{G} \overset{(7.6)}{=} k_7\bar{F}-k_6\bar{I}\bar{G}-k_8\bar{G}-\mu_G\bar{G}$$
$$= 0.022*26.17281262518254-0.06*8.837498959143081*0.8517150076840816$$
$$-0.14*0.8517150076840816-0.0058*0.8517150076840816=0$$

$$\dot{H} \overset{(7.7)}{=} k_8 G - \mu_H H$$
$$= 0.14 * 0.8517150076840816 - 0.00096 * 124.20843862052891 = 0$$

$$\dot{I} \overset{(7.8)}{=} k_9 \bar{D} + k_{10} \bar{L} \bar{J} - k_{11} \bar{I} + k_7 \bar{F} - k_6 \bar{I} \bar{G}$$
$$= 0.14 * 0.11976548876967165 + 0.1 * 4.027063329945519 * 0.9667161628812981$$
$$- 0.06 * 8.837498959143081 + 0.022 * 26.17281262518254$$
$$- 0.06 * 8.837498959143081 * 0.8517150076840816 = 0$$

$$\dot{J} \overset{(7.9)}{=} k_{11} \bar{I} - k_{10} \bar{L} \bar{J} - k_{12} \bar{J} - \mu_J \bar{J}$$
$$= 0.06 * 8.837498959143081 - 0.1 * 4.027063329945519 * 0.9667161628812981$$
$$- 0.14 * 0.9667161628812981 - 0.0058 * 0.9667161628812981 = 0$$

$$\dot{K} \overset{(7.10)}{=} k_{12} \bar{J} - \mu_K \bar{K}$$
$$= 0.14 * 0.9667161628812981 - 0.00096 * 140.97944042012693 = 0$$

$$\dot{L} \overset{(7.11)}{=} k_{11} \bar{I} - k_{13} \bar{L} - k_{10} \bar{L} \bar{J}$$
$$= 0.06 * 8.837498959143081 - 0.035 * 4.027063329945519$$
$$- 0.1 * 4.027063329945519 * 0.9667161628812981 = 0$$

■

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